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Multi-index analysis for wildfire severity and vegetation dieback assessment in semi-arid forest ecosystems

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Wildfires are among the most serious disturbances affecting semi-arid Mediterranean forest ecosystems, where abrupt fire-induced damage may coexist with gradual vegetation dieback driven by drought, soil degradation, and other environmental stressors. This study assessed wildfire impacts and vegetation decline in the Ouled Yagoub State Forest (Khenchela Province, northeastern Algeria) using a multi-index remote sensing approach. Cloud-free Landsat 8 OLI and Sentinel-2 images acquired between 2019 and 2022 were processed in Google Earth Engine to analyze vegetation dynamics before, during, and after the major 2021 wildfire event. Ten spectral indices related to greenness, burn severity, canopy water status, pigment stress, and soil background effects were derived and interpreted within three complementary themes: vegetation dynamics, burned-area severity, and forest dieback. The results showed that, according to the NDVI-based classification, 38.04% of the massif fell within high to very high wildfire-risk classes, 40.62% within the moderate class, and 20.33% within low to very low classes. Burn severity analysis based on dNBR for 2020–2021 indicated that 30.98% of the study area was affected by high to very high severity, 42.74% by moderate severity, and 26.28% by low to very low severity. The 2021–2022 dNBR comparison further showed that 75.23% of the area remained within high to very high disturbance classes, whereas only 3.01% fell into low to very low classes, confirming limited short-term recovery after the fire. NBR and dNBR were the most effective indices for delineating burned areas and severity gradients, while SAVI, MSAVI, and OSAVI improved the identification of vegetation loss and soil exposure in sparsely vegetated zones. In contrast, GNDVI, SIPI, and NDWI provided useful information on gradual physiological stress and dieback beyond the most severely burned sectors. Overall, the combined use of fire-sensitive, soil-adjusted, and stress-related indices provided a more robust interpretation of post-disturbance forest dynamics than any single index alone, offering a practical tool for post-fire assessment, ecological monitoring, and restoration planning in semi-arid Mediterranean forests.

KEYWORDS

burn severity mapping, forest dieback, Google Earth Engine, post-fire recovery, semi-arid Mediterranean forest, spectral indices

1 Introduction

Wildfires represent one of the most significant natural disturbances affecting forest ecosystems worldwide, generating profound ecological, social, and economic consequences. Large wildfire events lead to extensive habitat destruction, alter ecosystem functioning, and release substantial amounts of greenhouse gases into the atmosphere, thereby contributing to climate change. In recent years, several catastrophic wildfire events have occurred across different regions of the world, particularly in North America, Australia, and the Amazon Basin. Notable examples include the boreal forest fires in Alaska and Siberia in 2019, the Amazon fires during August–September 2019, the California wildfires in October 2019, and the large-scale fires that devastated southeastern Australia between December 2019 and January 2020 (Gajendiran et al., 2024). These events clearly demonstrate that wildfire occurrence results from complex interactions between climatic conditions, ecological characteristics, and human activities (Gajendiran et al., 2024) .

Extreme weather conditions such as prolonged droughts, heatwaves, and strong winds play a crucial role in intensifying wildfire behavior and facilitating rapid fire spread. Previous studies have shown that a relatively small number of large wildfire events account for the majority of the total burned area, a pattern that is particularly pronounced in Mediterranean ecosystems (Gajendiran et al., 2024). Arid and semi-arid regions are especially vulnerable to wildfire hazards due to the combination of high temperatures, low precipitation, and dense, highly flammable vegetation. These conditions are common in the Mediterranean Basin, North Africa, and parts of the Middle East, where wildfires frequently cause severe ecological degradation and threaten biodiversity (Bowman et al., 2017).In addition to climatic drivers, human activities significantly contribute to wildfire occurrence and propagation. Land-use change, agricultural expansion, uncontrolled burning, and urban development in forested areas increase the likelihood of fire ignition and complicate fire management efforts. The expansion of urban settlements into forested areas, commonly referred to as the wildland–urban interface (WUI), has further increased fire risk in many regions (Bowman et al., 2017). Consequently, wildfire monitoring and risk assessment have become essential components of sustainable forest management and ecosystem conservation strategies (Bowman et al., 2017).

Algeria, located within the Mediterranean and Saharan Atlas biogeographic regions, is particularly exposed to wildfire risk. Algerian forests have experienced an increasing frequency of wildfire events over the past decades, with large burned areas reported in regions such as the Aurès Mountains, Kabylie, and the Tell Atlas (Megrerouche, 2006). The Aurès Mountains in eastern Algeria represent an important ecological zone that acts as a natural barrier against desertification and supports diverse forest ecosystems (Beghami et al., 2012). Within this region, the Ouled Yagoub State Forest, covering approximately 27,305 hectares, represents a critical ecological and socio-economic resource. However, this forest ecosystem is increasingly threatened by wildfire activity due to prolonged summer droughts, rising temperatures, and growing human pressures such as uncontrolled grazing, agricultural expansion, and tourism activities (Chafai et al., 2023). Therefore, effective monitoring strategies are required to better understand vegetation dynamics and wildfire impacts in this vulnerable forest ecosystem (Chafai et al., 2023).

Remote sensing technologies provide powerful tools for monitoring forest ecosystems and assessing wildfire impacts across large spatial scales. Satellite-derived spectral indices enable the quantification of vegetation structure, physiological condition, and ecosystem disturbances. Among the most widely used indices, the Normalized Difference Vegetation Index (NDVI) assesses vegetation greenness and photosynthetic activity, while the Normalized Burn Ratio (NBR) is effective for detecting burned areas and evaluating burn severity. In addition, the Structure Insensitive Pigment Index (SIPI) provides valuable information regarding pigment ratios associated with vegetation stress (Chen et al., 2024).

Other spectral indices, including the Enhanced Vegetation Index (EVI), Green NDVI (GNDVI), and soil-adjusted vegetation indices such as SAVI, OSAVI, and MSAVI, improve vegetation monitoring by reducing soil background and atmospheric effects while enhancing sensitivity to vegetation structure, chlorophyll concentration, and canopy water content. These indices are particularly useful in Mediterranean and semi-arid ecosystems where vegetation stress responses are complex and often difficult to interpret (Allen et al., 2015; Bowman et al., 2017). Multi-scale optical remote sensing datasets have also been successfully used to monitor Mediterranean oak forests and assess vegetation responses to environmental variability and ecosystem disturbances (Adeline et al., 2024).

Recent advances in remote sensing have integrated multispectral satellite data with machine learning and spatial modeling techniques to improve wildfire risk mapping and vegetation monitoring. For example, combining Sentinel-2 multispectral imagery with hybrid machine learning algorithms has significantly improved forest fire risk prediction and spatial susceptibility mapping (Zerouali et al., 2025). Similarly, cloud-based platforms such as Google Earth Engine enable large-scale spatiotemporal analysis of land use and land cover dynamics using Random Forest (Bihon et al., 2025). Integrating vegetation indices with machine learning has also improved estimation of vegetation structural parameters such as Leaf Area Index (LAI) (Chatterjee et al., 2025).

Despite the widespread use of spectral indices in wildfire monitoring, limitations remain in accurately distinguishing between abrupt vegetation damage caused by wildfires and gradual forest dieback caused by drought, soil degradation, or disease. In Mediterranean forest ecosystems, these processes often produce similar spectral responses in satellite imagery, making their differentiation particularly challenging. Moreover, most studies focus either on wildfire mapping or general vegetation health monitoring without integrating multiple spectral indices to detect different disturbance mechanisms simultaneously (Mishra et al., 2024).

Satellite-based remote sensing has become an essential tool for monitoring vegetation dynamics and forest ecosystem functioning at multiple spatial and temporal scales. In Mediterranean environments, multi-scale remote sensing datasets have been successfully used to monitor oak forest ecosystems and assess vegetation responses to climatic variability and environmental stress (Adeline et al., 2024). These approaches highlight the potential of optical remote sensing for improving our understanding of Mediterranean forest dynamics and disturbance processes.

Recent studies have also explored the integration of multispectral imagery with machine learning algorithms to improve vegetation monitoring and biophysical parameter estimation. For example, (Chatterjee et al., 2025) demonstrated that combining vegetation indices derived from multispectral imagery with machine learning techniques significantly improves the estimation of vegetation structural parameters such as Leaf Area Index (LAI). These findings highlight the growing importance of integrating spectral indices and advanced analytical approaches for improved ecosystem monitoring.

The novelty of this study lies in combining ten widely used spectral indices—NDVI, NBR, dNBR, NDWI, SIPI, EVI, GNDVI, MSAVI, SAVI, and OSAVI—derived from Landsat-8 imagery for 2019–2022 to analyze vegetation responses in the Ouled Yagoub State Forest. We hypothesized that fire-sensitive indices (especially NBR and dNBR) would respond more strongly to abrupt wildfire damage, whereas chlorophyll- and pigment-related indices (such as GNDVI, GCI, and SIPI) would better capture progressive vegetation dieback and The objectives of this research are to:

- Evaluate the effectiveness of multispectral vegetation indices for wildfire severity mapping in a semi-arid Mediterranean forest ecosystem.
- Analyze vegetation decline and post-fire recovery dynamics using multi-temporal satellite imagery.
- Develop a multi-index analytical framework capable of distinguishing between abrupt wildfire-induced vegetation damage and gradual vegetation dieback.

The results of this study aim to improve operational forest monitoring, support wildfire management decision-making, and contribute to sustainable forest conservation strategies in Algeria.

2 Materials and methods

2.1 Location of the study area

The Ouled Yagoub forest massif, covering approximately 27,305 ha in the Aurès Mountains (Khenchela Province, Algeria), is located in the north-western part of the province. Ras Tafer (Unit n°07) constitutes the main entrance to the State Forest (CFK, 2023). The massif is bounded by the Remila plain and the national road connecting Khenchela to Batna to the north, Beni Melloul forest to the south, Beni Oudjana forest and Mellagou plain to the west, and Douar Ouled Bouderham and Douar Ouled N'sigha to the east (CFK, 2023; Figure 1).

The massif is situated within a semi-arid to sub-humid transition zone. Its altitudinal range extends from 869 m in the northern valley to 2,171 m in the highest peaks, with pine forests occupying 1,000–1,400 m and cedar forests 1,400–2,171 m (CFK, 2023). The terrain is characterized by rugged topography, including southern peaks such as Guern El Kebch (1,413 m), Djebel El Oud (1,560 m), and Djebel Aurès (1,521 m), and eastern and central peaks of the cedar forests including Djebel Chentgouma (2,113 m), Djebel Beker (2,080 m), Djebel Bezaz (1,941 m), Djebel Aidel (2,173 m), and Djebel Feraoun (2,093 m) (Bentouati, 2006).

2.2 Description of the study area

The region is dominated by forest and shrubland vegetation. The climate is marked by hot, dry summers and cold winters, favoring wildfire occurrence.

2.2.1 Geology

The pine forests develop on Upper Cretaceous deposits within the Djebel Aur'es syncline, characterized by marly, marl-limestone, and limestone facies. The cedar forests occur on Lower Cretaceous deposits with marly-limestone, sandy, and dolomitic facies, with Triassic outcrops along fault zones.

2.2.2 Pedology

Soils are generally shallow and skeletal (¡20 cm) on steep slopes, while flatter areas present relatively deeper soils supporting a well-developed herbaceous layer. Lithosols limit herbaceous establishment, leading to severe erosion in some zones. Humus layers are poor, composed of pine needles, holm oak leaves, and other shrub species. Calcimorphic soils of the rendzina type are also present (Bentouati, 2006; Figure 2).

2.2.3 Hydrographical network

The massif is drained by numerous dense talwegs forming steep, entrenched streams, with temporary regimes prone to high flow, particularly on southern slopes (Chafai et al., 2023). Major northern wadis include Oued Khafadja, O. Bou Barou, O. Boustane, O. Issoual, O. Ibikane, O. Tafrent, O. Tamarsit, and O. Tizougas; southern wadis include O. Tamza, O. El Houdh, O. Chentgouma, O. Zarif, O. Aziza, O. Maazouz, O. Taguerjoumt, O. Houira, and

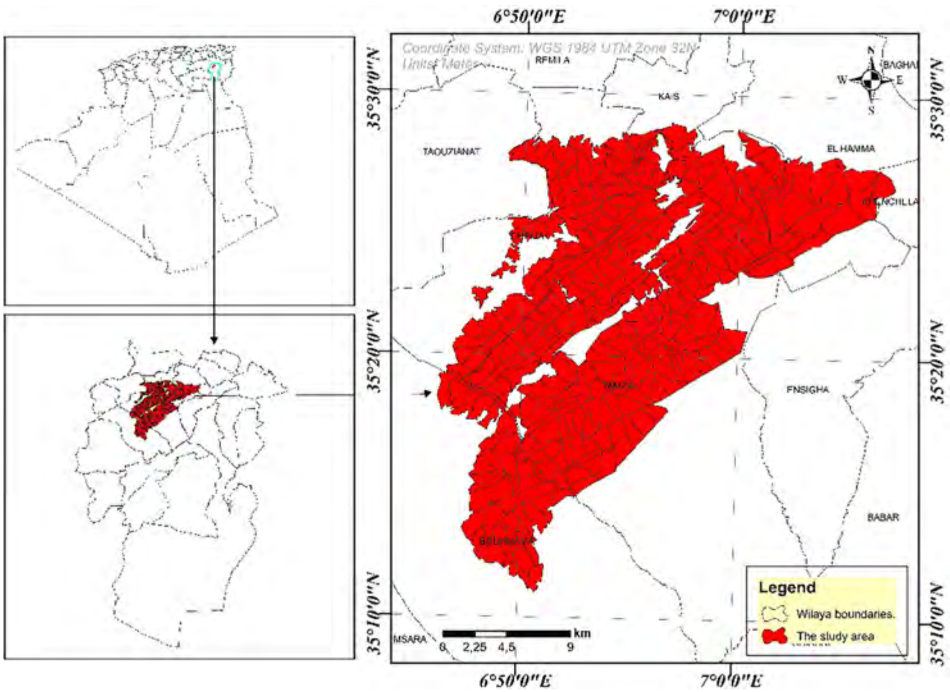


FIGURE 1
Geographical location of the Ouled Yagoub State Forest within Khenchela Province, northeastern Algeria.

Ighzer Iguechlane (SMH, 2020). Several springs supply water, such as El-Ansel.

2.2.4 Climate

The Ouled Yagoub massif has a continental, semi-arid climate with hot summers (up to 40°C) and cold winters (−2°C in cedar forests; 5°C in pine forests), though a sub-humid microclimate occurs in cedar areas (SMH, 2020). Temperature: Average annual temperatures range from 1.9°C in February to 35.7°C in August. Precipitation: Mean annual precipitation is 450.6 mm; March is the wettest month (61.5 mm), July the driest (18.2 mm). Ombrothermic Diagram Emberger Climagram: The area is classified as semi-arid with cool winters. Other parameters: Winter-dominant winds and hot summer sirocco winds facilitate fire spread; mean annual wind speed is 6.3 m/s, max 8 m/s (SMH, 2020). Humidity ranges from 40.4

2.2.4.1 Temperature and precipitation

We specified the monthly temperature extremes and averages based on data from the El-Hamma meteorological station (2011–2020), highlighting February as the coldest month (1.9°C) and August as the hottest (35.7°C) (Boumaiza, 2012).

We clarified precipitation patterns, noting an annual mean of 450.6 mm, with March as the wettest month (61.5 mm) and July as the driest (18.2 mm) (Boumaiza, 2012; Table 1).

2.2.4.2 Climate synthesis

2.2.4.2.1 Ombrothermic diagram of Bagnauls and Gaussen

The ombrothermic diagram of Bagnouls and Gaussen allows the identification of dry and wet periods by analyzing monthly precipitation and average monthly temperatures (Dajoz, 2003). The ombrothermic diagram of Ouled Yagoub Forest indicates two distinct periods:

Wet period: divided into two phases: the first from January to May and the second from September to December.

Dry period: extends from June to August (Figure 4).

2.2.4.2.2 Emberger climagram

The Emberger climagram, defined by Stewart (1968), is calculated using the formula:

Where: $Q2 = 3,43\, P / (M - m)$

- PPP = mean annual precipitation (mm)
- MMM = average maximum temperature of the hottest month (°C)
- mmm = average minimum temperature of the coldest month (°C)
- 3.43 = Stewart coefficient established for Algeria

Using climatic data from 2011 to 2020:

- P = 450.6 mm
- M = 35.7 °C
- m = 1.9 °C

The Emberger quotient (Q2) is 45.73, classifying the Ouled Yagoub State Forest within the semi-arid bioclimatic stage with cool winters.

2.2.5 Flora and fauna

The Ouled Yagoub massif covers 27,305 ha, with: Pine forests: 24,000 ha, average age 70 years. Cedar forests: 3,000 ha, located in

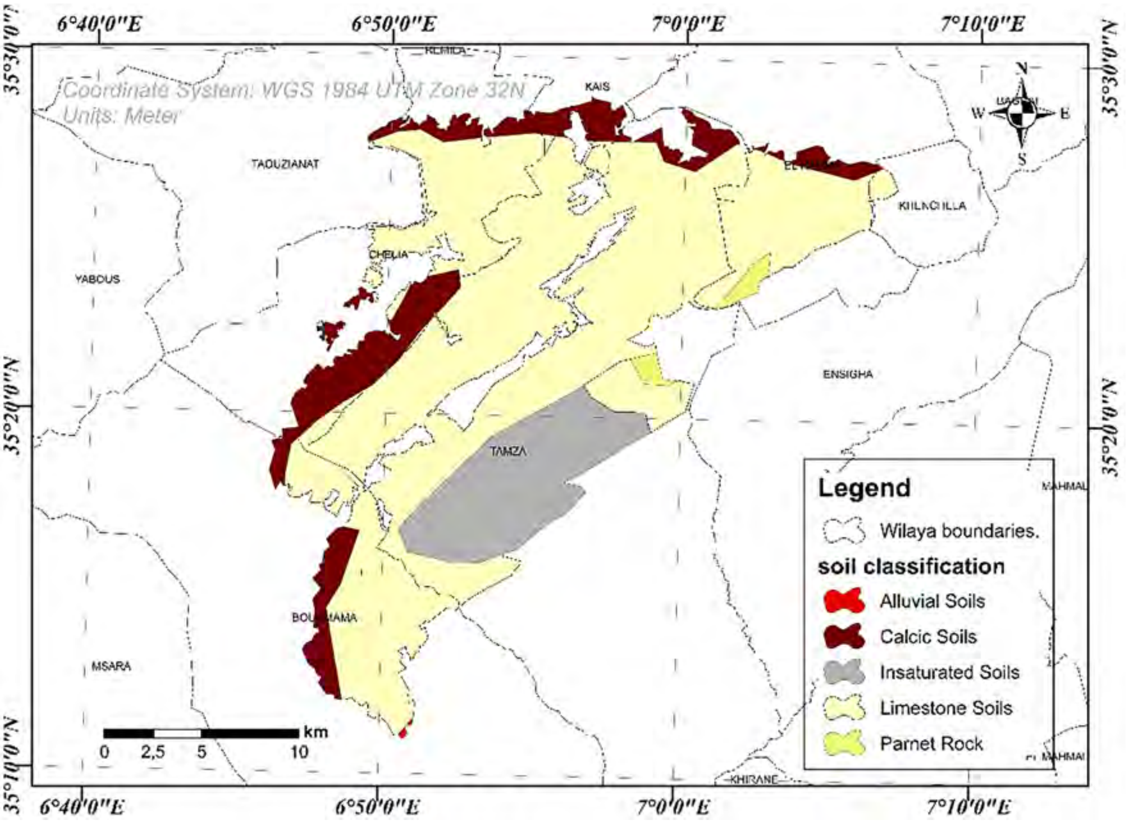


FIGURE 2
Soil classification of the Ouled Yagoub State Forest within Khenchela Province, northeastern Algeria.

humid northeastern zones, with holm oak coppices in both units (CFK, 2023).

Understory species: *Diss, Filaire*, *Rosemary, Globularia, Juniper, Maple, Fraxinus dimorpha, Alfa* (CFK, 2023), Fauna:

- Insects: *Thaumetopoea bonjeani*
- Mammals: *Sus scrofa barbarus, Canis aureus*
- Birds: *Picus viridis, Parus caeruleus*
- Reptiles: *Testudo graeca* (CFK, 2023).

Mapping fire-prone areas is a critical step in assessing wildfire risk and forms the foundation for implementing effective prevention strategies. This process relies on the integration of remote sensing data and GIS analyses. In our methodological approach, digital mapping utilizes specialized software for creating, analyzing, and updating spatial data and maps. These tools facilitate the visualization and interpretation of geographic information, enabling precise and detailed spatial analyses (Longley et al., 2015). By leveraging GIS, highly vulnerable areas can be identified through the combination of various remote sensing datasets, including vegetation indices and soil moisture (Parvar et al., 2024).

Remote sensing and geographic information systems (GIS) have become essential tools in forest fire management, enabling extensive monitoring and facilitating rapid responses to fire threats (Sivrikaya and Demirel, 2025). These technologies offer detailed insights into the factors that make certain areas prone to wildfires, including dry vegetation, fuel accumulation, and climatic variations (Chen et al., 2024).

2.2.6 Forest formations

According to Bneder (2008), the Ouled Yacoub State Forest (18,671 ha) consists of 81% genuine forests and 19% shrublands and wooded shrublands. The area of genuine forest covers 15,253 hectares and is distributed as follows:

- 3,926 ha of old Atlas cedar (Cedrus atlantica) stands, of which 3,049 ha are dense.
- 1,181 ha of old Atlas cedar stands mixed with holm oak (Quercus ilex), of which 933 ha are open.
- 638 ha of dense old Atlas cedar stands mixed with holm oak and dimorphic ash (Fraxinus dimorpha).
- 1,179 ha of holm oak forest, including 196 ha of open coppice, 735 ha of old open stands, and 248 ha of dense coppice.
- 993 ha of managed open Aleppo pine (Pinus halepensis) stands.
- 216 ha of young open Aleppo pine stands.
- 610 ha of open Aleppo pine forest at the perchis stage.
- 1,314 ha of managed dense Aleppo pine stands.
- 1,328 ha of young dense Aleppo pine stands.
- 1,849 ha of dense Aleppo pine forest at the perchis stage.
- 270 ha of old dense Aleppo pine stands.
- 1,684 ha of young open Aleppo pine and holm oak stands.
- 65 ha of burned Aleppo pine forest.

TABLE 1 Distribution of wildfires in the study area for the period 2011–2022 (CFK, 2023).

Year	Number of fire outbreaks	Burned area (ha)	Burned species	Intervention duration (h)	Weather conditions	Fire-fighting agencies
2011	4	1.15	PA	55:30	Very hot, sirocco	SF, PC, EAPC
2012	9	165.055	PA, CV, C, GO	258:35	Very hot, sirocco	SF, PC, EAPC, DW
2013	11	15.16	A, PA, B, CV, GO	96:20	Hot, sirocco	SF, PC, EAPC, PV
2014	12	173.385	PA, CV, GO	297:17	Hot, sirocco	SF, PC
2015	2	2.03	A, PA, CV	9:30	Dry and hot	SF, PC, EMPS
2016	6	7.61	CV, PA, GO, A	25:30	Dry and hot	SF, PC
2017	20	112.424	PA, CV, GO, A, F, D	831:31	Hot and moderate wind	SF, PC, EMPS, EAPC
2018	3	0.072	PA, GO, CV, A	11:28	Hot and seasonal wind	SF, PC, EAPC
2019	14	20.67	D, F, CV, PA, C	246:15	Hot and moderate wind	SF, PC, Workers
2020	6	7.05	PA, CV, GO, C, D	95:40	Hot and strong wind	SF, PC, GGR
2021	9	6422.558	PA, CV, C, A, D, F, GO	466:50	Hot and strong wind	SF, PC, GGR, CIT
2022	2	0.055	CV, PA, GO, A	9:25	Very hot and moderate wind	SF, PC, DW
Total	98	6927.219	/	/	/	/

PA, Aleppo pine (Pinus halepensis); CV, Holm oak (Quercus ilex); GO, Juniperus oxycedrus; A, Atlas cedar (Cedrus atlantica); C, Cork oak (Quercus suber); D, Ash (Fraxinus spp.); F, Other forest species; CY, Cypress. Report source: BM, Brigade de Montage; PV, Police de la Forêt; PC, Centre de Commandement; CIT, Citizen Report; SF, Forest Service; EAPC, Firefighting and Protection Agency; DW, DFCI Watch; GGR, General Firefighting Group.

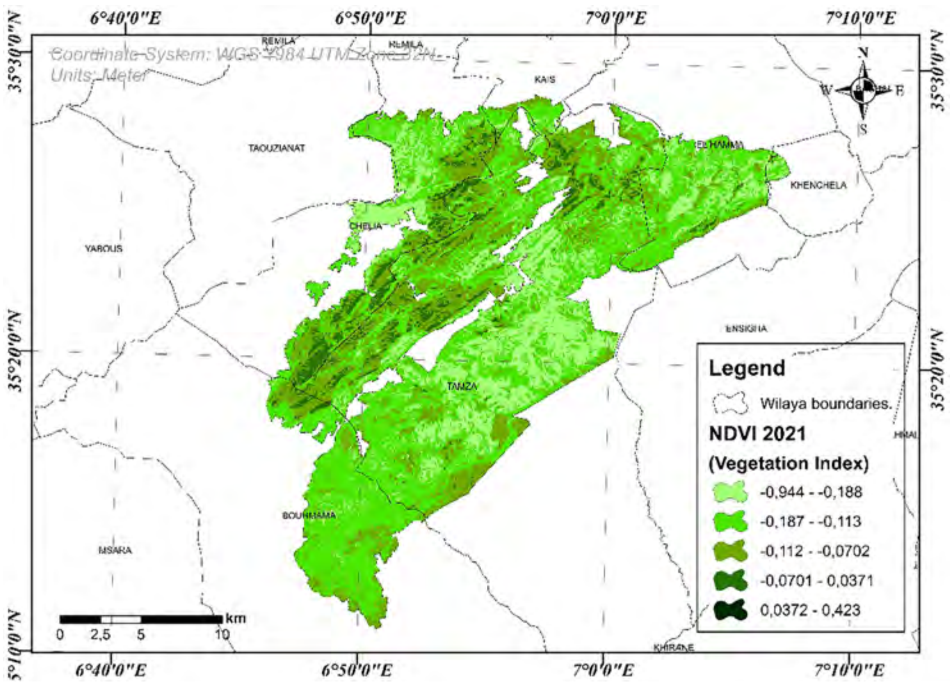


FIGURE 3
Normalized Difference Vegetation Index (NDVI).

Shrublands and wooded shrublands cover 3,293 ha, including:

- 1,141 ha of open shrublands dominated by holm oak and prickly juniper (Juniperus oxycedrus).
- 1,053 ha of dense shrublands dominated by holm oak and prickly juniper.
- 123 ha of dense shrublands dominated by holm oak, prickly juniper, and ash.
- 976 ha of wooded shrublands dominated by holm oak.

The area of reforestation accounts for 125 ha at the dense pole stage (Bneder, 2008).

The forest distribution map highlights the predominance of wooded formations in the Ouled Yacoub massif and confirms the major ecological importance of this forest complex within the Khenchela region. According to Bneder (2008), the massif covers

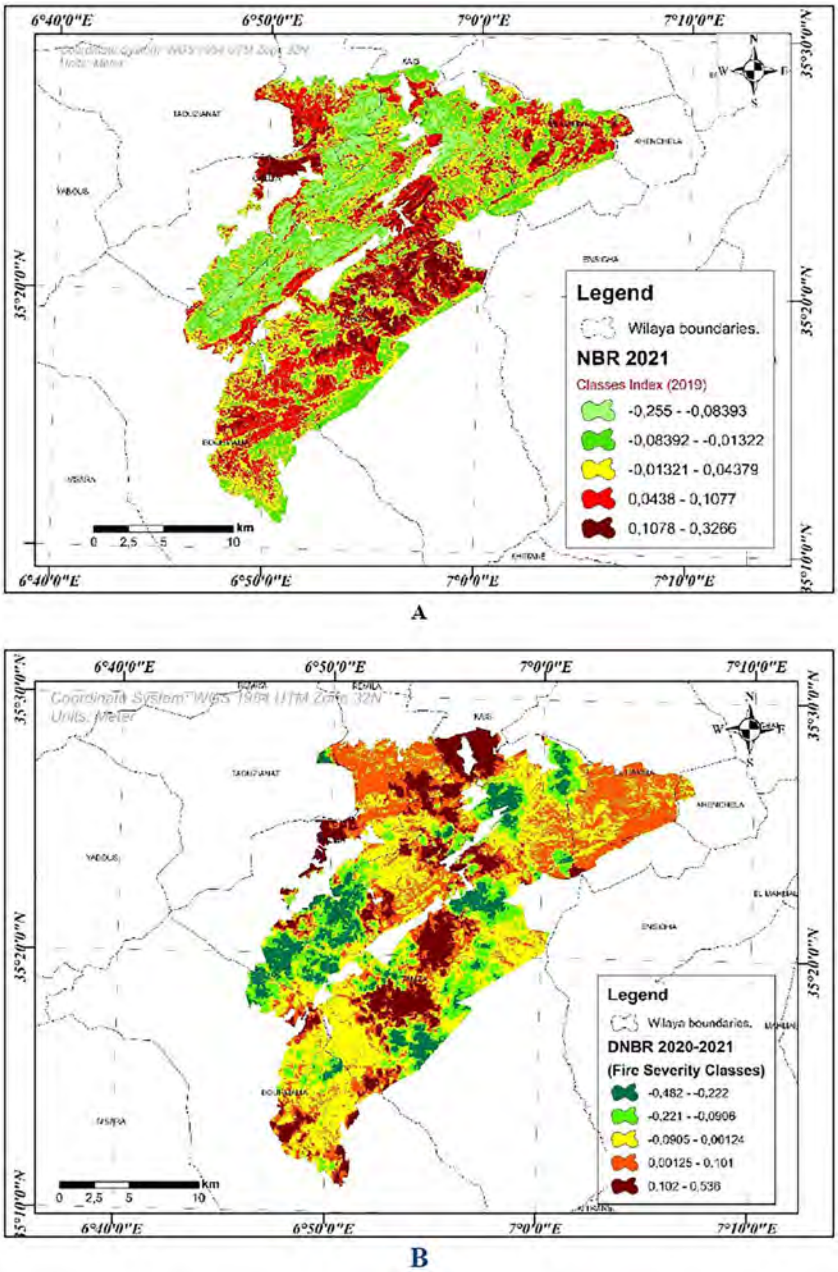


FIGURE 4
(A) Differenced normalized burn ratio (dNBR 2020–2021) and **(B)** Normalized Burn Ratio (NBR). **(B)** Differenced normalized burn ratio (dNBR 2020–2021).

18,671 ha, of which 81% consists of genuine forests and 19% of shrublands and wooded shrublands. The mapped pattern reflects this dominance of forest cover, with large and relatively continuous formations concentrated in the mountainous sectors, while more fragmented and open vegetation appears at the periphery. The forest structure is largely composed of Aleppo pine, Atlas cedar, and holm oak stands, distributed according to topographic and ecological gradients. Aleppo pine occupies extensive areas, whereas Atlas cedar is mainly restricted to higher and more humid zones, often mixed with holm oak. Shrublands and wooded shrublands dominated by holm oak and prickly juniper contribute to the heterogeneous and locally fragmented structure of the massif. This spatial organization indicates a diversified forest ecosystem with varying degrees of density, continuity, and ecological resilience, which is highly relevant for assessing wildfire sensitivity, vegetation degradation, and post-disturbance dynamics.

2.2.7 History of wildfires in the Ouled Yagoub State Forest

The Ouled Yagoub State Forest, like most Algerian forests, has suffered severe damage leading to the degradation of its forest ecosystem. This situation is attributed to several factors; including land clearing, soil erosion, grazing, and wildfires (CFK, 2023).

Wildfire data were obtained from the CFK in the form of annual reports covering an 11-year period (2011–2022), as well as reports

TABLE 2 Functional relationship among the selected spectral indices.

Index	Type of information	General relationship with the other indices	More sensitive to
NBR	Fire severity/post-fire condition	Often varies in the same direction as NDVI, NDWI, SAVI, MSAVI, and OSAVI after disturbance	Wildfire
NDVI	General vegetation vigor	Positively related to GNDVI, GCI, SAVI, MSAVI, and OSAVI	General vegetation condition
NDWI	Water content / hydric stress	Often related to NDVI and NBR; decreases strongly in burned or very dry areas	Wildfire + water stress
GNDVI	Chlorophyll activity	Positively related to NDVI and GCI; more useful for progressive stress	Forest dieback
GCI	Chlorophyll content	Closely related to GNDVI; decreases under progressive dieback	Forest dieback
MSAVI	Vegetation cover with soil correction	Positively related to SAVI, OSAVI, and NDVI; useful in sparsely vegetated areas	Wildfire in open environments
OSAVI	Vegetation with reduced soil effect	Very close to SAVI and MSAVI; positively related to NDVI	Wildfire + sparse vegetation
SAVI	Soil-adjusted vegetation cover	Very close to MSAVI and OSAVI; related to NDVI	Wildfire + sparse vegetation
SIPI	Pigment stress	Inversely related to GNDVI and GCI in dieback-affected areas	Forest dieback

on forest infrastructure, equipment, and the DFCI (Defense and Forest Fire Control) system.

Supplementry Appendix Table 1 presents the distribution of wildfires according to various parameters (date, time, location, etc.) and provides a basis for analyzing and interpreting the impacts caused by these fires.

2.3 Remote sensing data, indices, and software

The study employed satellite imagery from the Landsat 8 Operational Land Imager (OLI) and Sentinel-2 sensors, with spatial resolutions of 30 and 10 m, respectively. Cloud-free images covering the years 2019–2022, corresponding to pre- and post-wildfire periods, were acquired and processed using the Google Earth Engine (GEE) platform. The datasets were obtained from the USGS Earth Explorer. Ten widely used vegetation indices were computed in GEE to assess vegetation responses, differentiate abrupt wildfire damage from gradual dieback, and identify fire-prone areas. The resulting spectral indices and raster datasets were exported for further spatial analysis and cartographic visualization, conducted using ArcGIS 10.8 and QGIS. Additional data layers, including a Digital Elevation Model (DEM) and basemaps from Google Earth, were incorporated to support a comprehensive assessment of wildfire risk and vegetation conditions. and satellite imagery, including Landsat data for 2020, 2021, and 2022 and Sentinel-2 imagery for 2021.

2.3.1 Satellite images were selected according to the following criteria

2.3.1.1 The images were selected based on the following criteria

- Cloud cover less than 10%
- Acquisition during the peak vegetation season (May–September) to ensure consistent vegetation signals
- Availability of images before and after wildfire events

All satellite data were processed using Google Earth Engine (GEE) and geographic information system (GIS) software (arcmap).

2.3.1.2 Preprocessing steps included

- Atmospheric correction using surface reflectance products
- Cloud masking to remove contaminated pixels
- Image clipping to the study area boundary
- Spatial harmonization between Landsat-8 (30 m) and Sentinel-2 (10–20 m)

These preprocessing procedures ensured consistent spectral information for calculating vegetation indices and conducting spatial analyses.

2.3.2 Spectral indices and analytical rationale

To evaluate wildfire severity, post-fire vegetation dynamics, and progressive vegetation stress, ten spectral indices were calculated from the multispectral imagery: NDVI, NBR, dNBR, NDWI, SIPI, EVI, GNDVI, MSAVI, SAVI, and OSAVI (see Table 2). These indices were selected because they provide complementary information on burned-area detection, vegetation greenness, canopy moisture, physiological stress, and soil-background effects in semi-arid environments.

The indices were grouped functionally as follows. NBR and dNBR were used as the principal fire-sensitive indices for identifying burned areas and estimating burn severity. NDVI, EVI, and GNDVI were used to assess vegetation greenness, vigor, and post-fire recovery dynamics. NDWI was used to characterize vegetation moisture status and water stress. SIPI was included to detect pigment-related physiological stress, which is useful for identifying gradual dieback processes. SAVI, MSAVI, and OSAVI were incorporated to reduce the effect of exposed soil and soil brightness, which is particularly important in sparsely vegetated semi-arid landscapes. The combined use of these indices was intended to improve the differentiation between abrupt wildfire damage and progressive vegetation decline.

2.4 Spectral indices

Several spectral indices were calculated to assess vegetation condition and fire severity (Gülci and Wing, 2025).

2.4.1 Normalized Difference Vegetation Index

The Normalized Difference Vegetation Index (NDVI) is one of the most widely used vegetation indices in ecological research, providing valuable information on vegetation health, plant cover, and photosynthetic activity (Viju and Nambiar, 2023).

NDVI is calculated as follows (Equation 1):

$$NDVI = NIR - Red \div NIR + Red \quad (1)$$

The Normalized Difference Vegetation Index (NDVI) is derived from the light reflected by vegetation, captured by satellite sensors in the red (RED) and near-infrared (NIR) spectral bands (Jackson and Huete, 1991). It is calculated using reflectance values from the red channel, around 0.66 µm, and the near-infrared channel, around 0.86 µm (Gao, 1996). NDVI quantifies the structure of the interior mesophyll of healthy green leaves, which strongly reflect NIR radiation, while leaf chlorophyll and other pigments absorb most of the visible (VIS) light. When vegetation is stressed or unhealthy, these reflectance patterns are altered, leading to changes in NDVI values (Dutta et al., 2015; Ghafarian Malamiri et al., 2018). NDVI values range from −1 to +1, with healthy vegetation typically between 0.2 and 0.8 (Gessesse and Melesse, 2019).

2.4.2 Normalized Burn Ratio

The Normalized Burn Ratio (NBR) is a widely used index for assessing fire severity and identifying areas at high risk. It is calculated from the near-infrared (NIR, Band 5) and shortwave infrared (SWIR2, Band 7) bands of pre-fire and post-fire satellite images, as expressed in (Equation 2) (Khallef, 2019):

$$NBR = NIR - SWIR \div NIR + SWIR \quad (2)$$

NBR values allow for distinguishing burned and unburned areas and provide a quantitative measure of fire intensity (Zennir and Khallef, 2023).

2.4.3 Differenced Normalized Burn Ratio

The Difference Normalized Burn Ratio (dNBR) quantifies burn severity by calculating the dif ference between pre-fire and post-fire NBR images (Ghazali et al., 2025) (Equation 3) :

$$dNBR = NBR_{pre} - NBR_{post} \quad (3)$$

2.4.4 Normalized Difference Water Index

The Normalized Difference Water Index (NDWI) was developed to evaluate vegetation water content and surface moisture conditions. When the shortwave infrared (SWIR) band is used instead of near-infrared (NIR), NDWI becomes highly sensitive to plant water stress (Chandrasekar et al., 2024) (Equation 4).

$$NDWI = NIR - SWIR \div NIR + SWIR \quad (4)$$

- Key points: NDWI is sensitive to vegetation and soil water content, often combined with shortwave infrared bands (SWIR) for plant water status assessment.
- Usage: Detecting flooded fields, irrigated lands, and wetlands.

2.4.5 Optimized Soil-Adjusted Vegetation Index

OSAVI is an optimized version of SAVI, using a fixed soil correction factor (Fern et al., 2018) (Equation 5).

$$OSAVI = NIR - Red \div NIR + Red + 0.16 \quad (5)$$

2.4.6 Modified Soil-Adjusted Vegetation Index

The Modified Soil-Adjusted Vegetation Index (MSAVI) is designed to minimize the influence of soil brightness on vegetation measurements, making it particularly suitable for areas with sparse vegetation or exposed soil, where NDVI may become saturated. MSAVI has been widely applied in semi-arid and low-vegetation regions as expressed in (Equation 6) (Qi et al., 1994).

$$MSAVI = 2 \times NIR + 1 - \sqrt{(2 \times NIR)^2 - 8(NIR - Red)} \div 2 \quad (6)$$

2.4.7 Green Normalized Difference Vegetation Index

GNDVI is a modification of NDVI that replaces the red band with the green band, improving sensitivity to chlorophyll concentration and nitrogen status (Equation 7) (Gao et al., 2024).

$$GNDVI = NIR - Green \div NIR + Green \quad (7)$$

2.4.8 Green Chlorophyll Index

The Green Chlorophyll Index (GCI) estimates chlorophyll concentration in vegetation based on green reflectance, and it is widely used as an indicator of plant health and photosynthetic activity. It has been calculated as follows (Gao et al., 2024) (Equation 8):

$$GCI = NIR \div Green - 1 \quad (8)$$

This index is particularly effective for detecting variations in chlorophyll content related to environmental stress, nutrient deficiency, and pesticide effects (Sun et al., 2025).

2.4.9 Normalized Difference Snow Index

The Normalized Difference Snow Index (NDSI) is used to detect snow-covered surfaces by contrasting reflectance in the green and shortwave infrared (SWIR1) spectral bands., the index calculated according to the following (Wang et al., 2022) Equation (9):

$$NDSI = Green - SWIR \div Green + SWIR \quad (9)$$

The Normalized Difference Snow Index (NDSI), developed by Hall et al. (2001), is one of the most effective image-based snow mapping methods, monitor seasonal snow dynamics, and distinguish snow from clouds and bright soil surfaces (Sibandze et al., 2014). This method takes advantage of the ratio of snow's strong absorption and high reflectance in the visible and short-wave infrared portions of the electromagnetic spectrum, respectively (Hall et al., 2001).

2.4.10 Structure Insensitive Pigment Index

SIPI estimates the ratio of carotenoids to chlorophyll (Equation 10), indicating early plant stress (Penuelas et al., 1995).

$$SIPI = NIR - Blue \div NIR - Red \tag{10}$$

2.4.11 Red-Edge Chlorophyll Index

The Red-Edge Chlorophyll Vegetation Index (ReCI) is sensitive to leaf chlorophyll content, which is strongly influenced by nitrogen availability. This index enables the monitoring of photosynthetic activity and the detection of variations in plant health. ReCI is calculated using the standard red-edge and near-infrared (NIR) spectral bands, as described in Equation 11 (Sun et al., 2025).

$$\mathrm{ReCI} = \frac{\mathrm{NIR}}{\mathrm{Red}} - 1 \tag{11}$$

2.4.12 Visible Atmospherically Resistant Index

VARI uses only visible bands and is suitable for RGB imagery as expressed in (Equation 12) (Gitelson et al., 2002).

$$VARI = Green - \frac{Red}{Green + Red - Blue} \tag{12}$$

2.4.13 Atmospherically Resistant Vegetation Index

The Atmospherically Resistant Vegetation Index (ARVI) is a vegetation index specifically designed to reduce the influence of atmospheric scattering effects, particularly those caused by aerosols and smoke. Unlike NDVI, which relies solely on the red and near-infrared bands, ARVI incorporates the blue band to correct for atmospheric contamination (Somvanshi and Kumari, 2020)and calculated as follows (Equation 13):

$$ARVI = \frac{NIR\left(2 * Red - Blue\right)}{NIR - (2 * Red - Blue)} \tag{13}$$

2.4.14 Enhanced Vegetation Index

EVI improves sensitivity in dense vegetation, reduces atmospheric, and soil influences (Somvanshi and Kumari, 2020) The Equation 14 was used for calculation of this index.

$$EVI = G * NIR - Red \div NIR + C_1 * Red - C_2 * Blue + L \tag{14}$$

G is the gain factor (typically 2.5).

2.5 Analytical workflow

The methodological workflow consisted of five main stages. First, the study area was delimited and described using forest-service data, environmental information, and ancillary geographic layers. Second, Landsat 8 and Sentinel-2 images were selected and processed in Google Earth Engine according to cloud, season, and wildfire-timing criteria. Third, the ten spectral indices were calculated for the selected years. Fourth, the indices were interpreted comparatively to distinguish wildfire-sensitive responses from vegetation-stress responses. In this framework, NBR and dNBR were prioritized for burn-severity mapping, while NDVI, EVI, GNDVI, and NDWI were used to analyze vegetation vigor, moisture status, and post-fire recovery, and SIPI, SAVI, MSAVI, and OSAVI were used to support the identification of progressive stress and soil-exposed disturbed surfaces. Fifth, all outputs were mapped and interpreted spatially in the GIS environment to compare burned sectors, stressed vegetation, and broader post-disturbance patterns within the massif. This logic is consistent with the intended multi-index framework of the manuscript.

2.6 Semi-quantitative validation and statistical framework

The statistical analysis was based on a comparative interpretation of the 2021 spectral-index class tables and on the functional relationships among the selected indices. This approach was adopted because a complete pixel-based extraction and field-validation dataset were not available. In recent remote-sensing studies, multi-index interpretation has been recommended when vegetation condition, burn severity, moisture stress, and pigment-related responses need to be assessed simultaneously (Yan et al., 2025).

The selected indices were grouped according to their ecological function. NBR and dNBR were considered fire-sensitive indices because they are strongly responsive to post-fire changes in near-infrared and shortwave-infrared reflectance and remain among the most widely used indicators for burned-area detection and burn-severity assessment (Guria et al., 2025; Lee et al., 2025). NDVI and EVI were used as general vegetation-vigor indicators, NDWI as a canopy-moisture indicator, GNDVI, GCI, and ReCI as chlorophyll-related indices, SAVI, MSAVI, and OSAVI as soil-adjusted vegetation indices, and SIPI as a pigment-stress indicator. The use of chlorophyll-sensitive indices is supported by recent reviews showing that vegetation stress and chlorophyll variation can be effectively captured by spectral variables in the visible and red-edge regions (Li et al., 2025).

A first step of the analysis consisted of organizing the map legends into comparative class tables. This standardization allowed low, intermediate, and high index values to be interpreted consistently across all maps. In this framework, lower NBR, NDVI, NDWI, SAVI, MSAVI, and OSAVI values were interpreted as indicators of abrupt disturbance, biomass loss, canopy desiccation, and increased soil exposure. By contrast, higher SIPI values and lower GNDVI and GCI values were interpreted as indicators of progressive pigment stress and chlorophyll degradation. This interpretation is consistent with the current understanding that vegetation indices differ in their sensitivity to canopy greenness, chlorophyll content, water status, and structural vegetation change (Yan et al., 2025).

A second step consisted of evaluating the expected functional relationships among indices (Table 2). NDVI, GNDVI, GCI, SAVI, MSAVI, and OSAVI were considered positively related because they all tend to decrease when vegetation vigor and canopy cover decline (Table 3). SAVI, MSAVI, and OSAVI were treated as closely related because they belong to the same family of soil-adjusted indices and are particularly useful in sparse semi-arid

TABLE 3 Expected correlation relationships among the selected spectral indices.

Index pairs	Type of relationship	Interpretation
NDVI—GNDVI	Strong positive	The more active the vegetation, the more both indices increase
NDVI—GCI	Strong positive	Overall vegetation vigor increases with chlorophyll content
NDVI—SAVI	Strong positive	Both indices describe vegetation cover
NDVI—MSAVI	Strong positive	Same logic, with soil-background correction
NDVI—OSAVI	Strong positive	Same logic, under semi-arid conditions
NBR—NDWI	Moderate to strong positive	Less burned areas tend to retain more moisture
NBR—NDVI	Moderate positive	Areas in better vegetation condition often show higher NBR values
GNDVI—GCI	Strong positive	Both indices reflect chlorophyll activity
SAVI—MSAVI	Very strong positive	Very similar indices, differing mainly in the degree of soil correction
SAVI—OSAVI	Very strong positive	Indices belonging to the same family
MSAVI—OSAVI	Very strong positive	Same expected behavior
SIPI—GNDVI	Negative	As pigment stress increases, chlorophyll activity decreases
SIPI—GCI	Negative	As SIPI increases, chlorophyll content tends to decrease
SIPI—NDVI	Moderate negative	Higher physiological stress is generally associated with lower vegetation vigor

TABLE 4 Spectral index classes used in the 2021 analysis.

Index	Class 1	Class 2	Class 3	Class 4	Class 5
NBR 2021	−0.255 to −0.08393	−0.08392 to −0.01322	−0.01321 to 0.04379	0.0438 to 0.1077	0.1078 to 0.3266
NDVI 2021	−0.944 to −0.188	−0.187 to −0.113	−0.112 to −0.0702	−0.0701 to 0.0371	0.0372 to 0.423
NDWI 2021	−0.447 to −0.241	−0.24 to −0.198	−0.197 to −0.165	−0.164 to −0.125	−0.124 to 0.0224
GNDVI 2021	−0.0224 to 0.119	0.12 to 0.16	0.161 to 0.193	0.194 to 0.235	0.236 to 0.447
GCI 2021	−0.0437 to 0.295	0.296 to 0.412	0.413 to 0.522	0.523 to 0.666	0.667 to 1.62
MSAVI 2021	−0.0322 to 0.167	0.168 to 0.229	0.23 to 0.288	0.289 to 0.35	0.351 to 0.654
OSAVI 2021	−0.0158 to 0.0983	0.0984 to 0.14	0.141 to 0.179	0.18 to 0.224	0.225 to 0.486
SAVI 2021	−0.0237 to 0.148	0.149 to 0.209	0.21 to 0.269	0.27 to 0.336	0.337 to 0.729
SIPI 2021	−4.315 to 1.259	1.26 to 1.65	1.651 to 2.139	2.14 to 5.463	5.464 to 20.62

vegetation, where soil background strongly affects reflectance. In contrast, SIPI was interpreted as inversely related to chlorophyll-sensitive indices, because pigment stress tends to increase as chlorophyll activity declines. This ecological complementarity among indices supports the use of a combined framework rather than an isolated index-by-index interpretation (Li et al., 2025; Yan et al., 2025).

The distinction between wildfire damage and forest dieback was then validated using an interpretation framework based on the dominant spectral behavior of each disturbance type. Wildfire was primarily associated with high dNBR, low NBR, low NDWI, and abrupt declines in vegetation-vigor indices, which together indicate combustion, sudden moisture loss, and structural vegetation damage. Forest dieback, on the other hand, was associated with elevated SIPI and lower GNDVI, GCI, and ReCI values, which reflect progressive physiological stress and chlorophyll degradation. Similar multi-index strategies have recently been highlighted as useful for improving post-fire interpretation and separating fire effects from broader vegetation-condition changes (Guria et al., 2025; Lee et al., 2025).

Because no systematic field dataset was available, the present validation should be considered a semi-quantitative ecological validation rather than a full inferential statistical validation. A stronger validation would require extracting representative pixel values from burned and dieback areas and then applying formal statistical tests. In such a case, normality should first be evaluated using the Shapiro–Wilk test, and group differences should then be assessed using either an independent-samples *t*-test or the Mann–Whitney U test, depending on data distribution (Shapiro and Wilk, 1965)).

Table 4 summarizes the ecological role of each selected index and its general relationship with the others. NBR is interpreted as a fire-sensitive index because it responds strongly to post-fire changes in near-infrared and shortwave-infrared reflectance. NDVI represents general vegetation vigor, while NDWI reflects canopy moisture and water stress. GNDVI and GCI are more closely associated with chlorophyll activity and are therefore more useful for identifying progressive vegetation decline. SAVI, MSAVI, and OSAVI belong to the same family of soil-adjusted indices and are especially valuable in semi-arid environments where bare soil influences the spectral signal. SIPI is distinct from the others because it is linked to pigment stress and often behaves inversely to chlorophyll-related indices in dieback-affected areas. This functional differentiation supports the methodological choice

TABLE 5 Functional role of each index in distinguishing wildfire and forest dieback.

Index	What it measures	Dominant response to wildfire	Dominant response to dieback	Usefulness for discrimination
NBR	Burned surface and vegetation condition	Very strong	Low to moderate	Excellent for wildfire detection
dNBR	Burn severity	Very strong	Very low	Best indicator of fire severity
NDVI	General greenness and biomass	Decreases after fire	Gradual decline	Good general indicator, but not specific
NDWI	Canopy moisture and water stress	Rapid decline after fire	Moderate decline under chronic stress	Very useful as a complementary index
EVI	Biomass and vigor in dense vegetation	Strong decline	Gradual decline	Good complement to NDVI
GNDVI	Chlorophyll activity and vegetation vigor	Localized decline	Diffuse and progressive decline	Useful for dieback detection
GCI	Chlorophyll content	Moderate	Strong sensitivity to progressive stress	Very useful for dieback
SIPI	Carotenoid/chlorophyll ratio	Low to moderate	Very strong	Very effective for dieback
SAVI	Vegetation with soil correction	Good response in open disturbed areas	Moderate response	Useful in sparsely vegetated areas
MSAVI	Vegetation with enhanced soil correction	Good response where soil is exposed	Moderate response	Useful in semi-arid environments
OSAVI	Vegetation with reduced soil effect	Good response to disturbed surfaces	Moderate response	Good complementary index
ReCI	Chlorophyll concentration	Moderate	Good	Useful for physiological stress
VARI	Visible canopy color	Low to moderate	Low to moderate	Limited specificity
NDSI	Snow or bright-surface detection	Not relevant here	Not relevant here	Should be excluded from the main analysis

TABLE 6 Classification of indices according to their analytical value.

Group	Indices	Recommended use
Fire-sensitive indices	NBR, dNBR	Burned area detection and fire severity mapping
Moisture and water-stress indices	NDWI	Identifying sudden or progressive canopy water loss
General vegetation-vigor indices	NDVI, EVI	Monitoring overall vegetation degradation and recovery
Dieback and physiological-stress indices	SIPI, GNDVI, GCI, ReCI	Detecting gradual stress and forest dieback
Soil-adjusted indices	SAVI, MSAVI, OSAVI	Improving analysis in sparsely vegetated or soil-exposed sectors
Indices of limited relevance here	NDSI, VARI	Use with caution or exclude from the core interpretation

of combining several indices rather than interpreting each map separately.

Table 5 presents the expected correlation structure among the indices. Strong positive relationships are expected among NDVI, GNDVI, GCI, SAVI, MSAVI, and OSAVI because these indices all reflect vegetation cover, vigor, or chlorophyll status, although with different sensitivity to soil background and canopy structure. NBR is also expected to correlate positively with NDWI and moderately with NDVI, because less-burned areas usually retain more moisture and healthier vegetation. In contrast, SIPI is expected to show negative relationships with GNDVI, GCI, and NDVI because pigment stress typically increases as chlorophyll activity and vegetation vigor decline. These expected relationships are ecologically coherent and provide an internal consistency check for the interpretation of the different maps.

Table 6 provides the five class intervals used to interpret the 2021 maps for each index. These classes form the basis of the comparative analysis because they standardize the distinction between lower and higher spectral responses. For NBR, lower values correspond to burned or strongly disturbed surfaces, whereas higher values indicate relatively healthier vegetation. For NDVI and NDWI, lower classes indicate reduced greenness and lower canopy moisture, respectively. For GNDVI and GCI, lower values reflect reduced chlorophyll activity and lower chlorophyll content. For SAVI, MSAVI, and OSAVI, lower classes indicate sparse vegetation and stronger soil influence. Finally, for SIPI, higher values indicate stronger pigment stress, making this index particularly useful for identifying progressive vegetation decline. By organizing the maps into comparable class ranges, Table 6 provides the practical basis for the semi-quantitative validation strategy used in this study

Table 7 is central to the methodological interpretation because it links each index directly to wildfire sensitivity, dieback sensitivity, and discriminatory value. NBR and dNBR are the most informative

TABLE 7 Relation de corrélation attendue entre les indices.

Paires d'indices	Type de relation attendue	Interprétation
NDVI—GNDVI	Positive forte	Plus la végétation est active, plus les deux indices augmentent
NDVI—GCI	Positive forte	La vigueur générale augmente avec le contenu chlorophyllien
NDVI—SAVI	Positive forte	Les deux décrivent le couvert végétal
NDVI—MSAVI	Positive forte	Même logique, avec correction du sol
NDVI—OSAVI	Positive forte	Même logique, en conditions semi-arides
NBR—NDWI	Positive modérée à forte	Les zones moins brûlées gardent plus d'humidité
NBR—NDVI	Positive modérée	Les zones à bon état végétal ont souvent un meilleur NBR
GNDVI—GCI	Positive forte	Les deux traduisent l'activité chlorophyllienne
SAVI—MSAVI	Positive très forte	Indices très proches, différant surtout par la correction du sol
SAVI—OSAVI	Positive très forte	Indices de même famille
MSAVI—OSAVI	Positive très forte	Même comportement attendu
SIPI—GNDVI	Négative	Quand le stress pigmentaire augmente, l'activité chlorophyllienne diminue
SIPI—GCI	Négative	Quand SIPI augmente, le contenu en chlorophylle tend à diminuer
SIPI—NDVI	Négative modérée	Un stress physiologique élevé s'accompagne souvent d'une baisse de vigueur

TABLE 8 Interpretation framework for distinguishing wildfire and forest dieback.

Disturbance type	Most relevant indices	Expected spectral response	Main interpretation
Wildfire	dNBR, NBR, NDWI, EVI	High dNBR, low NBR, low NDWI, abrupt drop in EVI	Sudden disturbance, biomass combustion, rapid moisture loss
Forest dieback	SIPI, GNDVI, GCI, ReCI, NDVI	High SIPI, low GNDVI/GCI, gradual decline in NDVI	Progressive stress, chlorophyll degradation, chronic water deficit
Mixed disturbance	NDVI, NDWI, SAVI, MSAVI, OSAVI	Intermediate values	Areas affected by both chronic stress and surface disturbance

TABLE 9 Annual wildfire statistics for the Ouled Yagoub State Forest over the period 2011–2022 (CFK, 2022).

Year	Number of fire outbreaks	Burned area (ha)	Burned species	Intervention duration (h)	Meteorological conditions	Firefighting agencies
2011	4	1.15	PA	55:30	Very hot, sirocco	SF, PC, EAPC
2012	9	165.055	PA, CV, C, GO	258:35	Very hot, sirocco	SF, PC, EAPC, DW
2013	11	15.16	A, PA, B, CV, GO	96:20	Hot, sirocco	SF, PC, EAPC, PV
2014	12	173.385	PA, CV, GO	297:17	Hot, sirocco	SF, PC

indices for wildfire because they respond very strongly to combustion and burn severity. NDWI complements them by detecting canopy moisture loss, while NDVI and EVI describe the overall decline in vegetation vigor after fire. In contrast, GNDVI, GCI, SIPI, and ReCI are more useful for identifying forest dieback because they are linked to chlorophyll activity, pigment stress, and progressive physiological decline rather than abrupt biomass loss. SAVI, MSAVI, and OSAVI are not fire-specific, but they improve the interpretation of disturbed areas where soil exposure is high. This table therefore demonstrates why the proposed framework is based on index complementarity: wildfire is best characterized by burn-sensitive and moisture-sensitive indices, whereas dieback is better detected by pigment- and chlorophyll-sensitive indices.

Table 8 reorganizes the indices into analytical groups, which improves methodological clarity. Fire-sensitive indices are used for burned-area detection and burn-severity mapping; moisture-related indices help identify abrupt or progressive canopy water loss; vegetation-vigor indices track overall degradation and recovery; dieback-related indices detect physiological stress; and soil-adjusted indices improve analysis in open or sparsely vegetated sectors. This grouping is important because it reduces confusion between indices that are broadly responsive to vegetation change and those that are more diagnostic of specific disturbance mechanisms. In practical terms, Table 8 supports the selection of a smaller, more coherent set of indices for interpretation and eventually for future statistical validation.

Table 9 refines the functional relationships shown in Table 5 by specifying the expected direction and strength of the correlations among the indices. The strongest positive relationships are expected within the vegetation-vigor and soil-adjusted groups, particularly among NDVI, SAVI, MSAVI, and OSAVI, and between GNDVI and GCI. The strongest negative relationships are expected between SIPI and the chlorophyll-related indices, because rising pigment stress usually accompanies declining chlorophyll activity. These expected correlations are not presented here as measured coefficients, but as an ecological hypothesis framework that can

TABLE 10 Monthly wildfire record in the Ouled Yagoub State Forest during 2021 (CFK, 2022).

Commune	Date and time	Fire location	Burned species and burned area (ha)
Tamza	25/06/2021, 10:05–11:40	Soumaat	0.02 (PA + CV)
Tamza	28/06/2021, 13:10–19:50	Bouhrawa	0.040 (PA)
Tamza	04/07/2021 to 13/07/2021, 11:20–11:40	Boughardhaiene	6141.438 (PA + C + F + GO + CY + CV)
Tamza	24/07/2021, 12:45–15:00	Tifighelanine	0.01 (PA)
Tamza	25/07/2021, 13:30–18:30	Tizi Nethelathe	0.050 (PA + CV + GO)
Tamza	01/08/2021, 15:00–21:00	Taghezout	1.000 (PA + CV + GO)
Tamza	02/08/2021 to 03/08/2021, 11:30–19:00	Ghil Oudjalile	80.000 (PA + CV + GO + A + F + D)
Tamza	04/08/2021 to 12/08/2021, 11:30–17:00	Ras Tafer	173.355 (PA + CV + GO + F)
El-Hamma	/	Ras Tafer	26.645
Total			6422.558

later be tested with pixel-based extraction and formal correlation analysis. This distinction is methodologically important because it prevents the overinterpretation of descriptive class-based patterns as if they were measured statistical coefficients.

Table 10 provides the final decision framework of the multi-index approach. Wildfire is defined by the combined presence of high dNBR, low NBR, low NDWI, and abrupt drops in EVI or other vigor-related indices, which together indicate combustion, sudden moisture loss, and structural vegetation damage. Forest dieback, by contrast, is defined by high SIPI, low GNDVI and GCI, and a more gradual decline in NDVI, reflecting physiological stress, pigment imbalance, and chronic water deficit. Mixed disturbance areas are expected to show intermediate values, especially for NDVI, NDWI, SAVI, MSAVI, and OSAVI, indicating the coexistence of surface disturbance and progressive vegetation decline. This table is the practical synthesis of the entire validation strategy, because it translates the spectral behavior of the indices into an operational ecological interpretation relevant for forest monitoring and management in semi-arid environments.

3 Results and discussion

This section presents the spatial distribution of spectral indices before and after the wildfire event. High burn severity was mainly observed in forested areas with dense pre-fire vegetation.

3.1 Spectral indices

In contrast, GNDVI and SIPI were more informative for identifying progressive vegetation stress outside the most severely burned areas. GNDVI highlighted more diffuse declines in chlorophyll-related activity, while SIPI was more sensitive to pigment imbalance linked to physiological stress. In the present case, elevated SIPI values in non-burned sectors suggest that some areas of the massif were already affected by vegetation stress unrelated to direct fire damage. This distinction is important because it supports the reviewer's point that abrupt wildfire impact and gradual dieback should not be interpreted as the same process.

The combined reading of dNBR + NDVI/EVI/NDWI + SIPI/GNDVI therefore provides a stronger analytical framework than any single index alone. Fire-sensitive indices identify the disturbance footprint, greenness and moisture indices describe post-fire vegetation condition, and pigment-sensitive indices help detect more progressive stress patterns. Recent studies also show that integrating multiple remotely sensed variables improves wildfire-related analysis and provides a stronger basis for advanced classification and predictive approaches (Guria et al., 2024; Mishra et al., 2024; Zerouali et al., 2025).

3.1.1 Normalized Difference Vegetation Index

In this study, five NDVI classes were defined to assess wildfire risk (Table 1). Based on the NDVI classification results presented in Table 1, the study area exhibits a significant wildfire risk profile. Over one-third of the massif (38.04%) is categorized as having high to very high risk, corresponding to NDVI values ranging from 0.1261 to 0.7942. This indicates that a substantial portion of the landscape consists of relatively healthy, dense vegetation, which can serve as abundant fuel during a fire. An additional 40.62% of the area falls into the moderate risk class (NDVI: −0.07423 to 0.126), representing transitional or stressed vegetation that still contributes considerable fuel load. In contrast, only 20.33% of the total area is classified as low to very low risk (NDVI: −0.4058 to −0.07422), typically associated with non-vegetated surfaces, bare soil, or severely degraded vegetation that presents minimal fire fuel.

According to Figure 3, NDVI values in the study area range approximately from −0.1 to 0.6. Areas with NDVI values below 0.2 dominate large portions of the massif, indicating sparse vegetation, bare soil, or severely degraded forest cover. Moderate vegetation conditions correspond to NDVI values between 0.2 and 0.4, while dense and healthy forest stands are characterized by NDVI values above 0.4, mainly on north-facing slopes. The spatial dominance of NDVI values < 0.3 confirms that a large part of the forest experienced strong vegetation degradation following the 2021 wildfire, with limited recovery observed in 2022.

3.1.1.1 Analysis of vegetation distribution

The NDVI map reveals the spatial distribution of vegetation and fire susceptibility: Areas with dense vegetation are primarily located on the north-eastern and southern exposures. Areas with

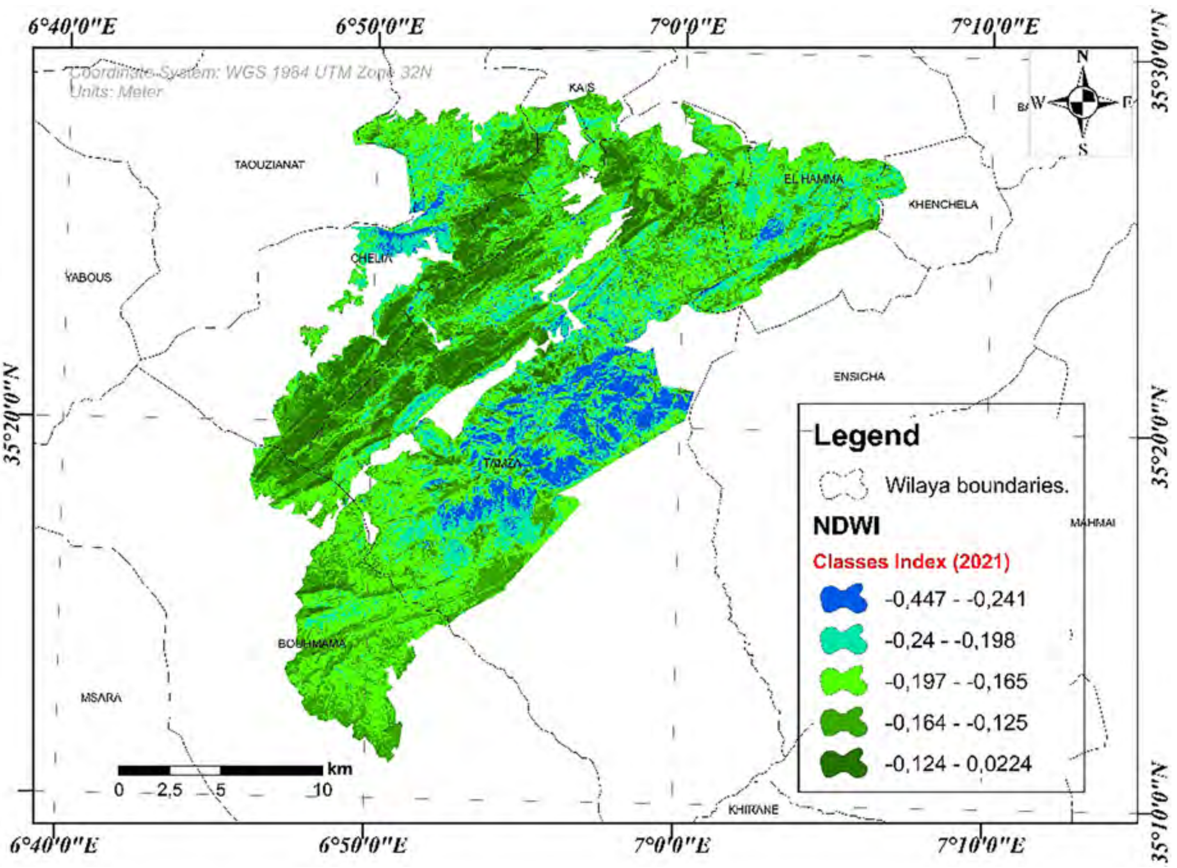


FIGURE 5
Normalized Difference Water Index (NDWI).

low or degraded vegetation are mostly found on south-eastern and south-western slopes. These patterns indicate that more than half of the forest massif exhibits low or absent vegetation cover, largely due to the major wildfire event in 2021, which caused significant canopy loss and long-term degradation.

3.1.2 Normalized Burn Ratio

According to the legend of Figure 4A (NBR 2021), Normalized Burn Ratio values are classified into five numerical classes:

- Areas characterized by negative NBR values (−0.255 to −0.01322) correspond to zones that experienced severe fire impact, where vegetation and surface organic matter were largely consumed. These values indicate a strong increase in SWIR reflectance following combustion, which is a well-known spectral signature of burned surfaces.
- Pixels with near-zero NBR values (−0.01321 to 0.04379) represent moderately affected areas, where partial vegetation loss and surface alteration occurred.

In contrast, areas exhibiting positive NBR values greater than 0.0438, particularly those within the 0.1078 to 0.3266 range, correspond to unburned or weakly affected vegetation, maintaining higher near-infrared reflectance and lower SWIR response.

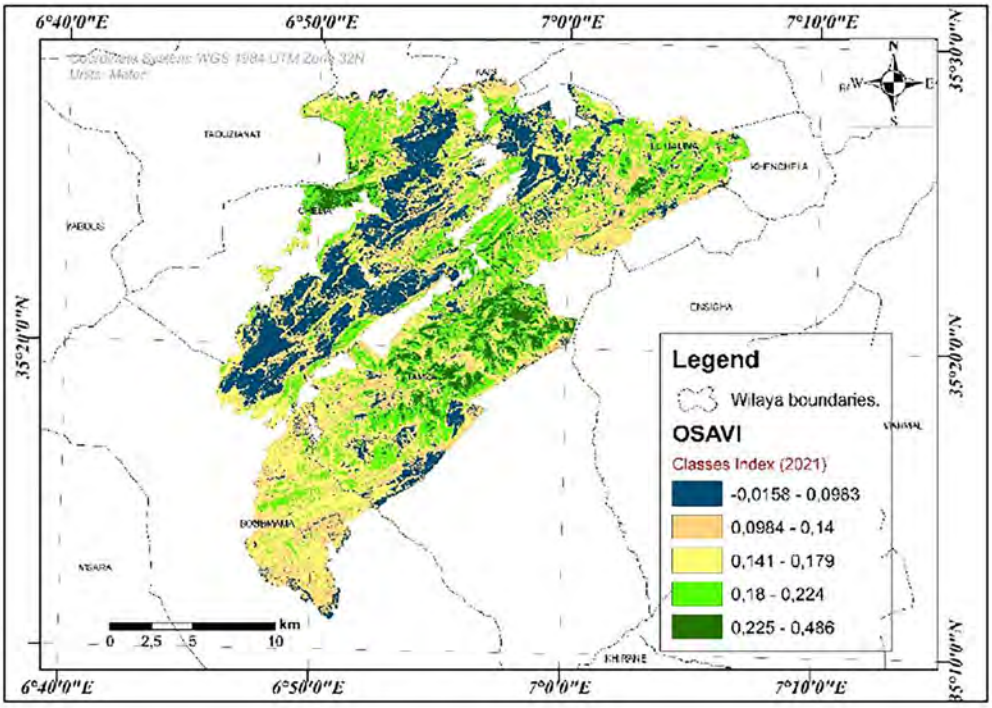
According to the NBR legend, areas with negative NBR values (−0.255 to −0.013) correspond to fire-affected zones, while positive values (> 0.0438) indicate unburned or weakly affected vegetation. The spatial dominance of negative NBR classes is consistent with the recorded burned area of 6,422.558 ha in 2021, confirming the effectiveness of NBR for wildfire delineation.

3.1.3 Differenced Normalized Burn Ratio

According to Figure 4B the dNBR map classifies fire severity into five classes, ranging from negative values (−0.482 to −0.0906) indicating unburned or vegetation regrowth areas, to positive values (0.00125 to 0.536) representing increasing fire severity. Areas with high positive dNBR values correspond to zones of strong vegetation loss and severe burning, while near-zero values indicate low to moderate fire impact.

3.1.4 Normalized Difference Water Index

Based on the classification provided in Figure 5, the Normalized Difference Water Index (NDWI) map illustrates a gradient of canopy water content or moisture stress across the study area for the year 2021. The highest values, depicted by the class ranging from −0.447 to −0.241, represent areas with the greatest relative vegetation water content, typically corresponding to healthy, well-hydrated plant canopies. The subsequent classes (−0.24 to −0.198 and 0.197 to 0.165) indicate moderate moisture levels,



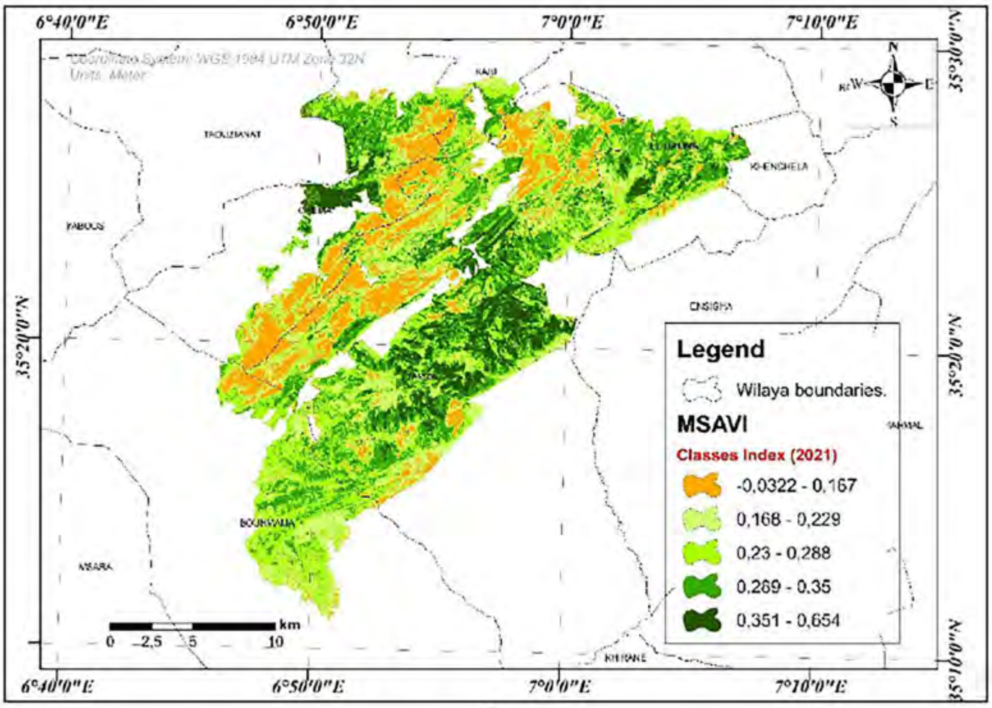


FIGURE 6
(A) Modified Soil-Adjusted Vegetation Index (MSAVI) and **(B)** Optimized Soil-Adjusted Vegetation Index (OSAVI).

often found in mixed or moderately stressed vegetation. In contrast, the lowest NDWI values, represented by the classes −0.164 to −0.125 and −0.124 to −0.024, signify areas of significantly lower canopy water content, which may correspond to dry, stressed, or sparse vegetation, as well as non-vegetated surfaces.

3.1.5 Optimized Soil-Adjusted Vegetation Index

According to Figure 6A, the Optimized Soil-Adjusted Vegetation Index (OSAVI) values in the Ouled Yagoub forest range from −0.0158 to 0.486 indicating strong spatial variability in vegetation density and soil influence. The lowest OSAVI

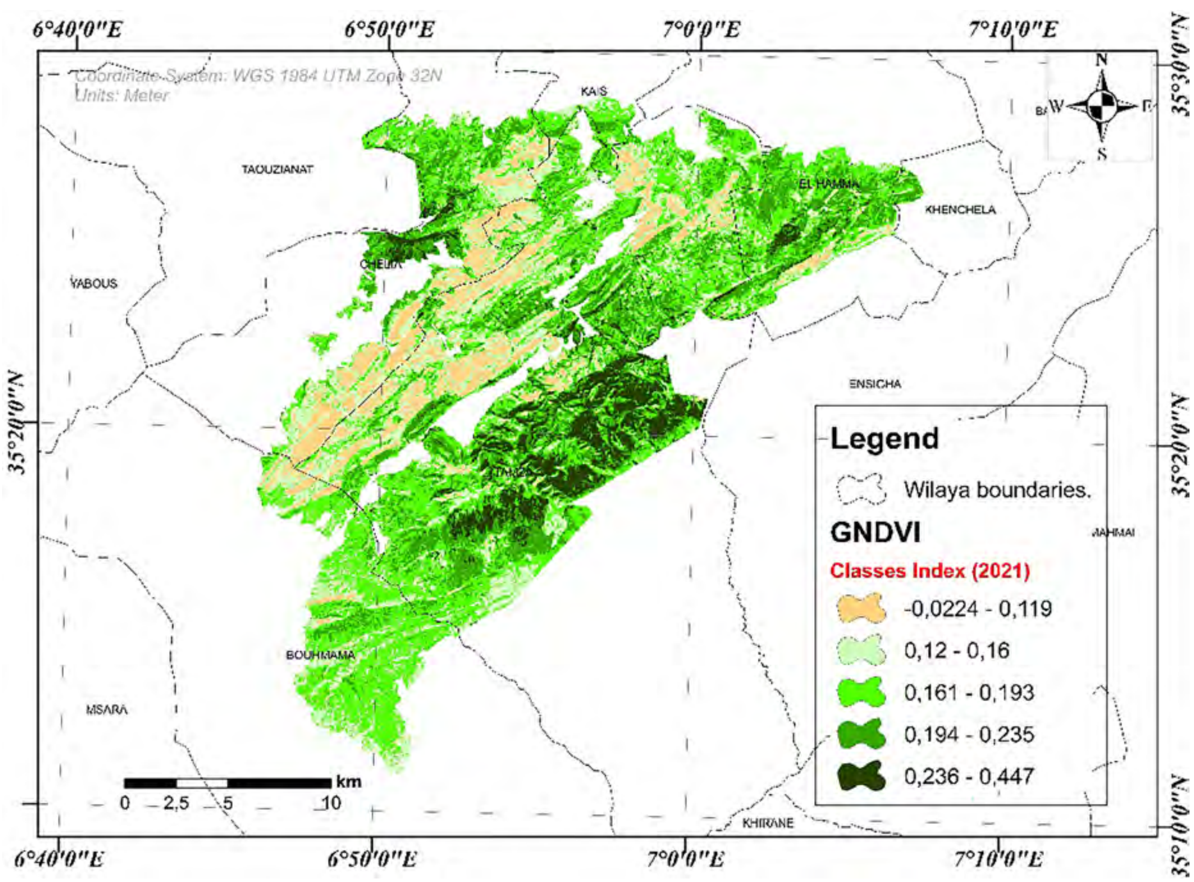


FIGURE 7
Green Normalized Difference Vegetation Index (GNDVI).

values, close to −0.0158, are mainly associated with severely disturbed areas, where vegetation cover is extremely sparse and soil reflectance dominates the spectral signal. These zones correspond to either recently burned areas or highly degraded surfaces characterized by exposed soil and reduced canopy cover. Intermediate OSAVI values represent areas with partial vegetation cover, reflecting moderate degradation consistent with forest dieback, where canopy density is reduced but not completely removed. The highest OSAVI values, approaching 0.486, are concentrated in dark blue zones and correspond to dense, healthy forest stands with limited soil background influence and high vegetation vigor. This spatial pattern confirms that OSAVI effectively captures the gradient from severely disturbed surfaces to intact forest, making it a robust indicator for distinguishing abrupt wildfire impacts from gradual forest decline in semi-arid Mediterranean environments.

3.1.6 Modified Soil-Adjusted Vegetation Index

According to Figure 6B, MSAVI values range from −0.0322 to 0.654, showing patterns similar to SAVI but with reduced soil background influence. Low MSAVI values (−0.0322–0.167) coincide with the severely burned zones identified by NBR, indicating extensive canopy loss and strong soil influence on spectral response. Intermediate MSAVI classes (0.168–0.229 and 0.23–0.288) dominate areas experiencing moderate vegetation degradation, likely associated with forest dieback or low-severity fire effects. High MSAVI values (0.351–0.654) correspond to preserved forest stands with dense vegetation cover.

3.1.7 Green Normalized Difference Vegetation Index

According to Figure 7, physiological stress indices reveal a contrasting pattern. GNDVI values (−0.0224 to 0.447) show widespread moderate declines not restricted to burned zones, indicating progressive chlorophyll degradation associated with forest dieback. Similarly, SIPI values range from −4.315 to 20.62, with elevated values (>2.14) predominantly occurring in non-burned areas, reflecting increased carotenoid-to-chlorophyll ratios typical of prolonged vegetation stress. In contrast, low SIPI values dominate burned areas, where pigments are largely destroyed by combustion.

3.1.8 Green Chlorophyll Index (GCI)

According to Figure 8A, The Green Chlorophyll Index (GCI) reveals marked spatial variability in chlorophyll content across the Ouled Yagoub forest. GCI values range from [-0.0437] to [1.06], with the lowest values predominantly observed in areas affected by forest dieback, indicating significant chlorophyll depletion and reduced photosynthetic capacity. In contrast, burned areas

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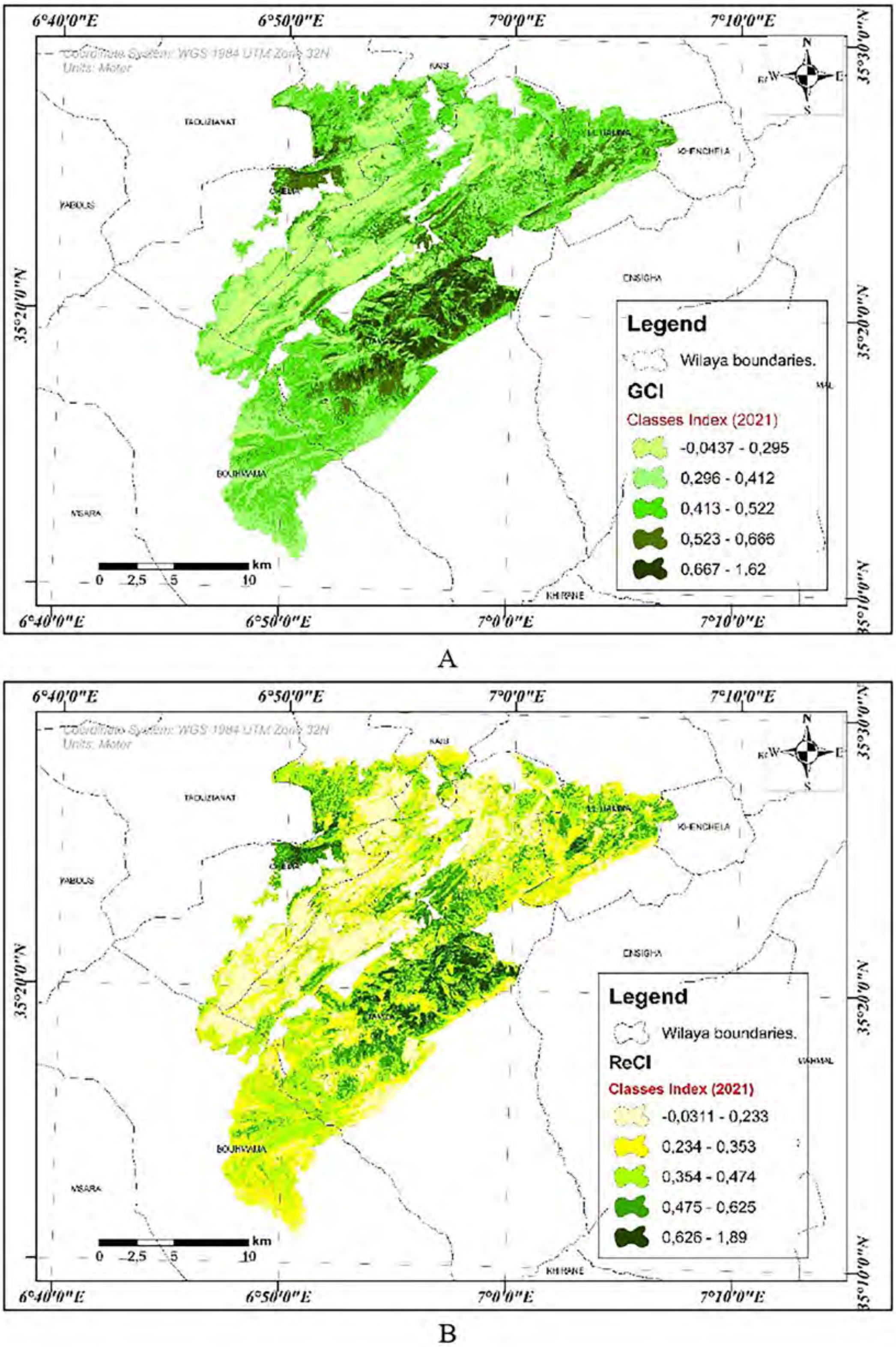

FIGURE 8
(A) Green Chlorophyll Index (GCI), and **(B)** Red-Edge Chlorophyll Index (ReCI).

exhibit uniformly low to moderate GCI values, reflecting the near-total destruction of leaf pigments following combustion rather than gradual physiological stress. Higher GCI values are mainly associated with unburned and structurally intact forest stands, corresponding to healthy vegetation with high chlorophyll concentration. This spatial pattern indicates that GCI is more sensitive to progressive chlorophyll degradation linked to forest dieback than to abrupt biomass loss caused by wildfire.

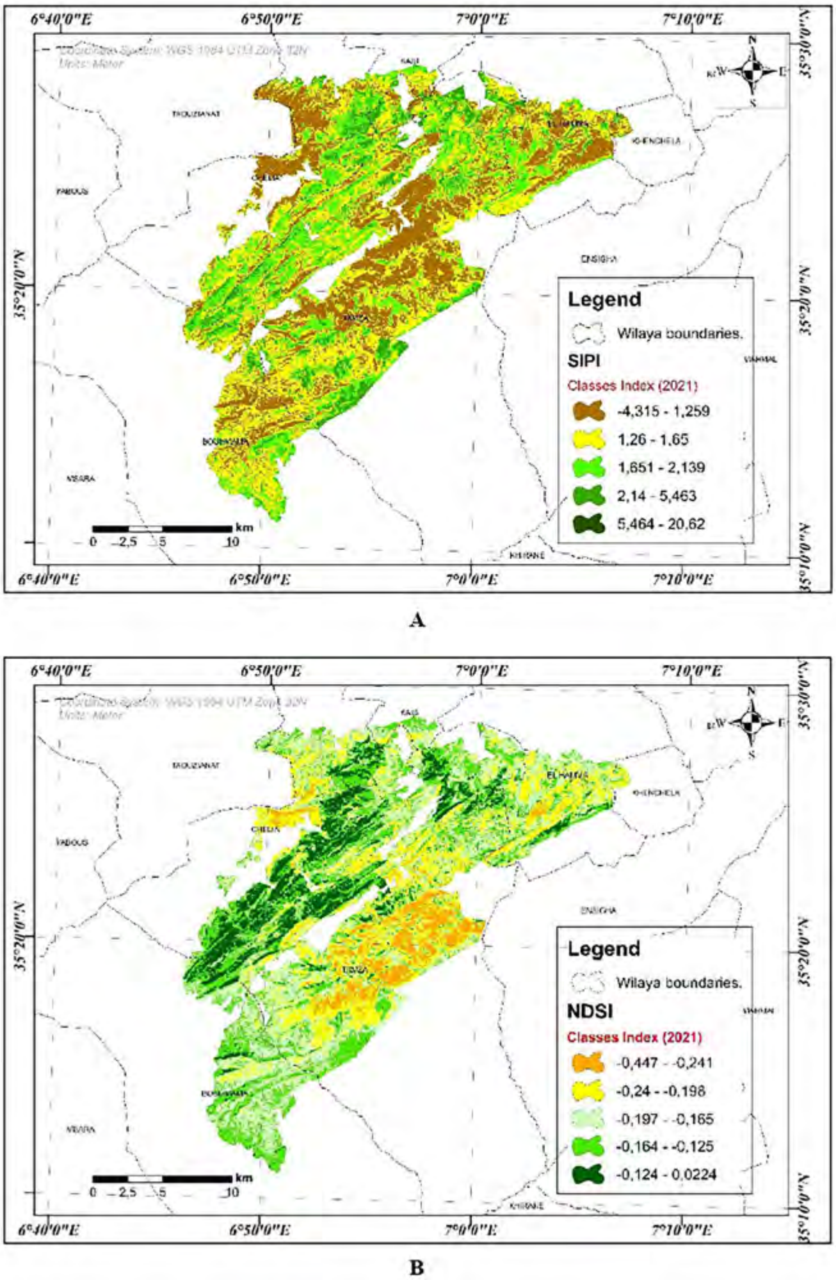

FIGURE 9
(A) Structure Insensitive Pigment Index (SIPI) and **(B)** Normalized Difference Snow Index (NDSI).

3.1.9 Red-Edge Chlorophyll Index

Based on the Red-Edge Chlorophyll Index (ReCI) map for 2021 presented in Figure 8B, the study area exhibits a pronounced gradient in vegetation chlorophyll content, which serves as a key indicator of plant health and photosynthetic activity. The lowest ReCI class, ranging from −0.0311 to 0.233, corresponds to areas with minimal chlorophyll, typically representing non-vegetated surfaces, bare soil, or severely stressed or senescent vegetation. Progressively higher index values reflect increasing chlorophyll density: the 0.234 to 0.353 and 0.354 to 0.474 classes indicate low to moderate chlorophyll content, often associated with sparse, degraded, or moderately healthy vegetation. The two highest classes, 0.475–0.625 and 0.626–1.89, identify regions of high to very high chlorophyll concentration. These areas represent dense, healthy, and actively photosynthesizing vegetation, which can correlate with higher biomass and potentially greater fuel availability for wildfires. Consequently, while these high-chlorophyll zones signify robust ecosystem productivity, they may also correspond to areas of elevated fire risk due to the abundance of combustible live fuel.

3.1.10 Structure Insensitive Pigment Index (SIPI)

According to Figure 9A, the SIPI map exhibits a wide range of values from −4.315 to 20.62, reflecting strong variability in vegetation pigment composition. Elevated SIPI values (2.14–5.463 and 5.464–20.62) are predominantly observed in areas not affected by wildfire, indicating an increased carotenoid-to-chlorophyll ratio typical of prolonged stress and early stages of forest dieback. In contrast, low SIPI values (−4.315–1.259) dominate the burned zones, where pigments have been largely destroyed by combustion. These results confirm the effectiveness of SIPI as an indicator of forest dieback rather than fire severity

3.1.11. Normalized Difference Snow Index

The Normalized Difference Snow Index (NDSI) values in the study area varied from −0.447 to 0.241 (Figure 9B). Positive NDSI values reflect surfaces with higher snow influence, while negative values are associated with snow-free land covers such as vegetation, bare soil, and water bodies. Intermediate values indicate mixed surface conditions and spatial heterogeneity. The resulting NDSI map demonstrates clear spatial variability, highlighting its effectiveness in representing snow cover distribution under varying environmental conditions.

3.1.12 Visible Atmospherically Resistant Index

According to Figure 10A, values (−0.237 to 0.0969) respond to both disturbance types, with strongly negative values in burned areas and slightly negative to near-zero values in dieback-affected zones. Although VARI captures changes in canopy color and structure, it shows limited specificity when used independently.

3.1.13 Atmospherically Resistant Vegetation Index

According to Figure 10B, ARVI values range from −0.1 to 0.32, with the lowest values (< -0.0038) spatially coinciding with burned areas, indicating abrupt biomass loss caused by wildfire. Moderate to high ARVI values characterize degraded and healthy vegetation, respectively, confirming the strong responsiveness of ARVI to fire-induced structural changes VARI

The present study provides a comprehensive assessment of wildfire impacts in the Ouled Yagoub State Forest using multi-temporal spectral indices derived from satellite imagery. The combined analysis of dNBR, NDVI, and related vegetation indices highlights the severity of the 2021 wildfire event and its profound effects on vegetation structure, ecosystem functioning, and post fire recovery dynamics. The integration of multiple spectral indices enabled a comprehensive assessment of wild- fire impacts and vegetation dynamics.

The results demonstrate that wildfire and forest dieback in the Ouled Yagoub forest generate distinct spectral responses that can be effectively discriminated using a multi-index approach. Wildfire impacts are most clearly detected by NBR, ARVI, and OSAVI, which highlight burned areas and severity gradients through abrupt vegetation loss and increased soil exposure. Soil-adjusted indices such as SAVI and MSAVI further refine detection in sparsely vegetated zones.

Forest dieback is better captured by indices sensitive to physiological stress and pigment composition, including GNDVI, SIPI, and GCI. These indices reveal progressive vegetation degradation not always coincident with burned areas, indicating stress-driven decline. Elevated SIPI and GCI values in non-burned zones signal early chlorophyll degradation, while NDWI and ReCI detect changes in canopy water content and pigment responses, providing complementary insight into subtle stress processes. NDVI and VARI offer additional context by reflecting reductions in greenness and changes in canopy color, but are insufficient alone to distinguish disturbance types. Overall, integrating fire-specific, soil-adjusted, and physiological stress indices provides a robust framework for differentiating wildfire effects from forest dieback. This multi-index approach enhances post-disturbance interpretation and supports effective monitoring and management in semi-arid Mediterranean forests.

3.2 Fire-related data processing

3.2.1 Analysis of wildfire records

The assessment of wildfire hazard was based on an exploratory statistical analysis of historical fire records. During the study period from 2011 to 2022, approximately 6,927.219 ha of forests, maquis, and shrublands were affected by fire, with a total of 98 wildfire outbreaks recorded (Table 10).

Based on the data presented in Table 10, it is evident that 2021 was an exceptional fire year, during which wildfires burned 6,422.558 ha, whereas the cumulative burned area for the remaining eleven years was only 504.661 ha. Over the entire study period, the mean burned area was approximately 50.46 ha year^{-1}, which can be classified as a small fire class. In contrast, the fires recorded in 2021 caused extremely severe damage, with an estimated burned area of 642.25 ha year^{-1}, corresponding to a very large fire class, according to the international classifications proposed by Teusan (1995), San-Miguel-Ayanz et al. (2013), and Tedim et al. (2013).

3.3 Exceptional case study: the major wildfire event of 2021

The summer of 2021 marked a major turning point for the Ouled Yagoub State Forest because of the severe wildfire events that affected its forest wealth and caused a substantial decline in its ecological, touristic, and economic value, with a total loss of 6422.558 ha (Table 11).

According to the monthly wildfire record reported by the Conservation des Forêts de Khenchela, the summer period (June, July, and August) recorded the highest number of fire events, generally occurring between 10:00 a.m. and 9:00 p.m. The burned areas affected several vegetation species, mainly Aleppo pine, holm oak, Atlas cedar, juniper, alfa grass, phillyrea, diss grass, and cypress.

Figure 11 shows that wildfire ignition points were distributed across multiple locations within the massif, favoring rapid and catastrophic fire propagation. Consequently, the total burned area reached 6422.558 ha.

3.3.1 Spatial distribution of wildfire ignition points

Figure 11 illustrates the spatial distribution of wildfire ignition points recorded during the 2021 fire season. Ignitions were dispersed across multiple locations within the massif, which

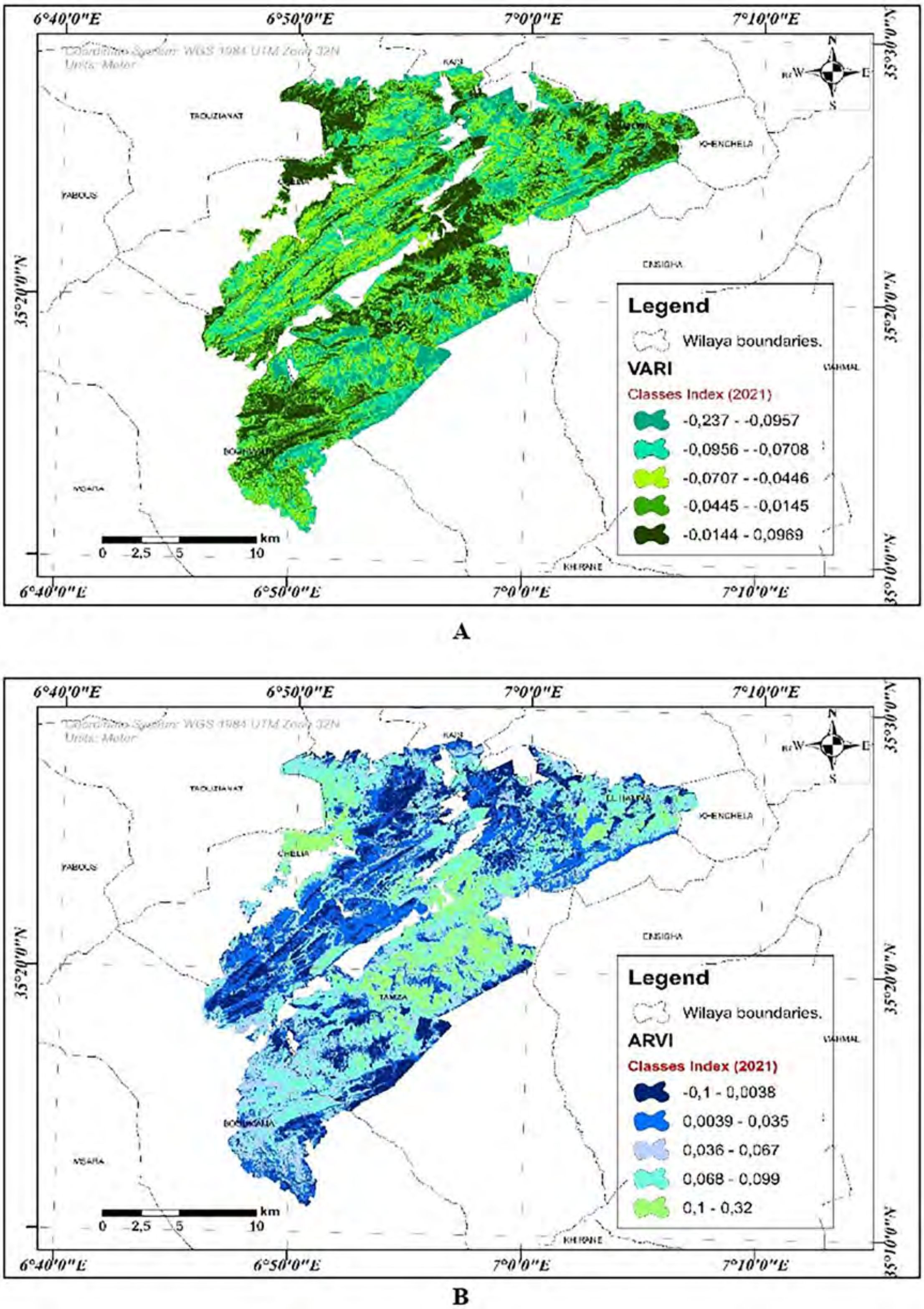

FIGURE 10
(A) Visible Atmospherically Resistant Index (VARI) and **(B)** Atmospherically Resistant Vegetation Index (ARVI).

facilitated rapid-fire propagation and contributed to the large spatial extent of the burned area. According to Figure 12, wildfire ignition points were spatially distributed across several locations within the massif, favoring a rapid and large-scale propagation of the fire. This spatial pattern significantly contributed to the severity of the event, resulting in a total burned area estimated at 6,422.558 hectares. Among the main factors influencing wildfire intensity, wind conditions played a critical role. At the

TABLE 11 NDVI classes and associated wildfire risk.

NDVI Classes (2022)	Area (ha)	Area (%)	Risk class
[-0.4058, −0.09519]	1,747.52	6.40	Very low
[-0.09518, −0.07422]	3,803.59	13.93	Low
[-0.07423, 0.126]	11,091.29	40.62	Moderate
[0.1261, 0.2389]	10,386.82	38.04	High
[0.239, 0.7942]	273.32	1.00	Very high

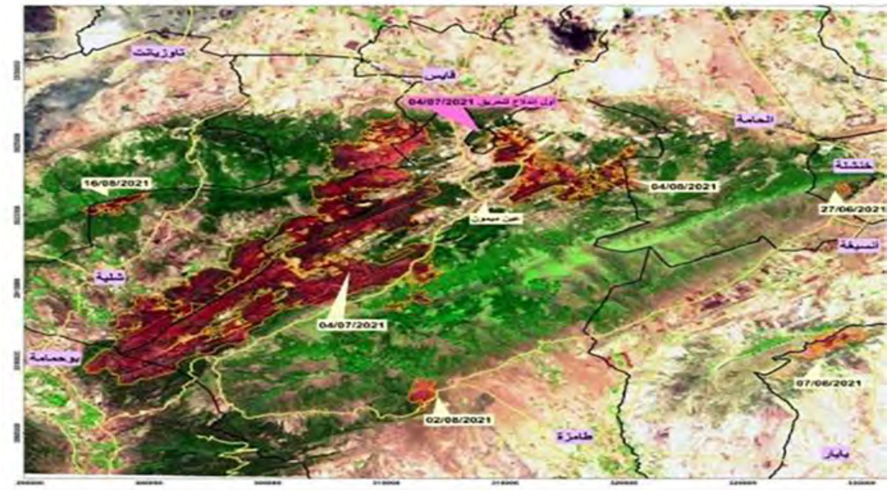

FIGURE 11
Map of the ignition points of the 2021 wildfires. Spatial distribution of wildfire ignition points recorded in 2021 in the Ouled Yagoub State Forest (CFK, 2022).

time of ignition, prevailing winds were predominantly from the south and southwest, directing the fire front toward the eastern and northern sectors of the massif (Figures 13, 14). This wind-driven fire behavior is typical of Mediterranean environments, where wind strongly governs both the rate and direction of fire spread (Gajendiran et al., 2024). In addition, elevated summer temperatures further intensified fire spread by lowering fuel moisture and increasing vegetation flammability, thereby enhancing combustion efficiency and overall fire intensity. The combined effect of adverse meteorological conditions and abundant fuel ultimately resulted in the rapid and uncontrolled expansion of the wildfire across the forested landscape (Filkov et al., 2023).

3.3.2 Meteorological conditions during the fire event

Meteorological data indicate that the wildfire spread occurred under adverse weather conditions. Prevailing winds originated mainly from the south and southwest, with wind speeds reaching up to 60 km/h (Figure 12). High summer temperatures were also recorded during the fire period. According to Figure 12, the wildfire spread rapidly due to adverse weather conditions, particularly high temperatures and wind speeds reaching up to 60 km/h. The combination of these conditions. Understanding the temperature and wind dynamics associated with a wildfire is essential for clarifying the event's scale and its potential long-term consequences for the forest ecosystem, the broader environment, and adjacent human settlements. These impacts encompass the tragic loss of human life, the widespread destruction of fauna and flora, a significant reduction in economic value, the degradation of air and water quality, and substantial damage to property and infrastructure. (Bowman et al., 2017). Consequently, implementing robust forest fire defense and control measures (DFCI), supported by appropriate silvicultural strategies, is essential to mitigate wildfire risk (Hassan et al., 2025). Key interventions must include public awareness and education campaigns, enhanced systems for early detection and continuous monitoring, the promotion of sustainable forest management practices, and the active restoration of burned areas to foster ecosystem recovery and resilience (FAO, 2006).

3.4 Burn severity mapping

Our results are consistent with recent studies showing that dNBR remains one of the most robust and operationally useful indicators for burn-severity assessment, especially when pre- and post-fire Sentinel-2 imagery are available (Gülci and Wing, 2025). Recent studies also confirmed the value of Sentinel-2-based approaches for wildfire mapping and post-fire assessment in heterogeneous environments (Suwanprasit and Shahnawaz, 2024). However, post-fire ecological condition cannot be fully interpreted from a single fire-sensitive index alone. A multi-index interpretation is more appropriate because moisture-related and vegetation-related spectral responses

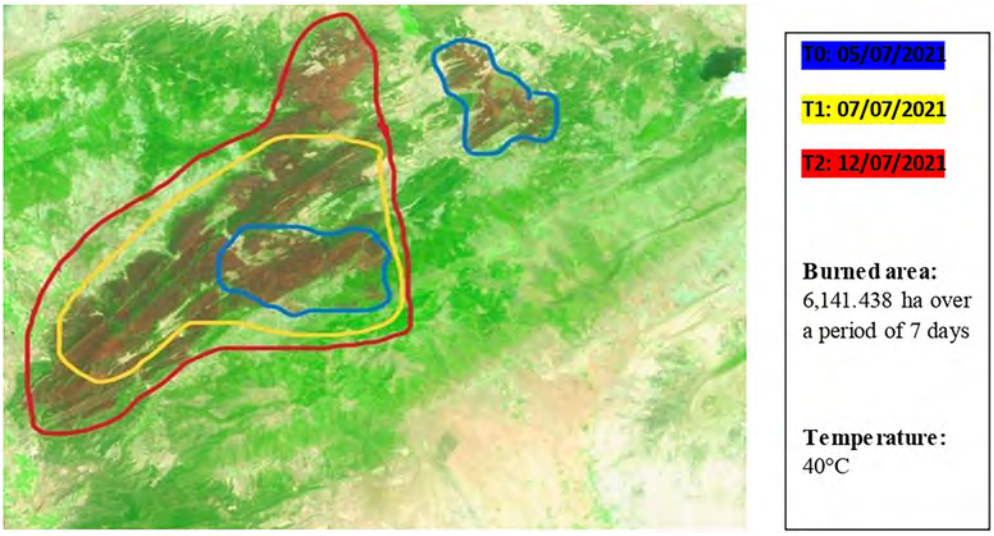

FIGURE 12
Wildfire propagation speed significantly amplified the spread and intensity of the fire (CFK, 2023).

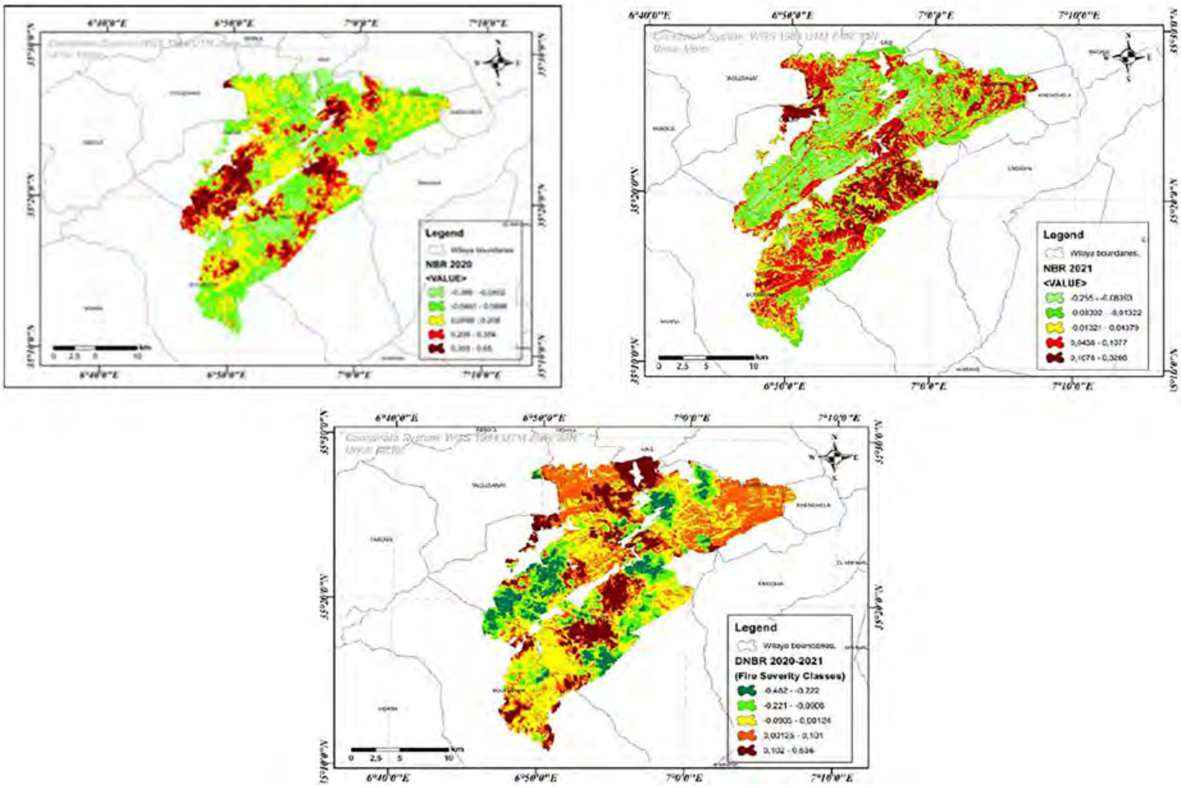

FIGURE 13
Spatial distribution of burn severity in the study area based on dNBR (2020–2021).

provide complementary information on canopy stress, surface condition, and early recovery dynamics after fire (Anees et al., 2025; Gülci and Wing, 2025). In this sense, our analytical framework supports a more ecologically meaningful interpretation of post-fire disturbance by combining fire-sensitive, vegetation-vigor, and moisture-related indices within a unified approach (Anees et al., 2025).

3.4.1 NBR and dNBR response to the 2021 fire event

The NBR and dNBR maps clearly delineated the areas most affected by the 2021 wildfire. Negative post-fire NBR values corresponded to sectors where vegetation and surface organic matter were strongly altered by combustion, while positive values

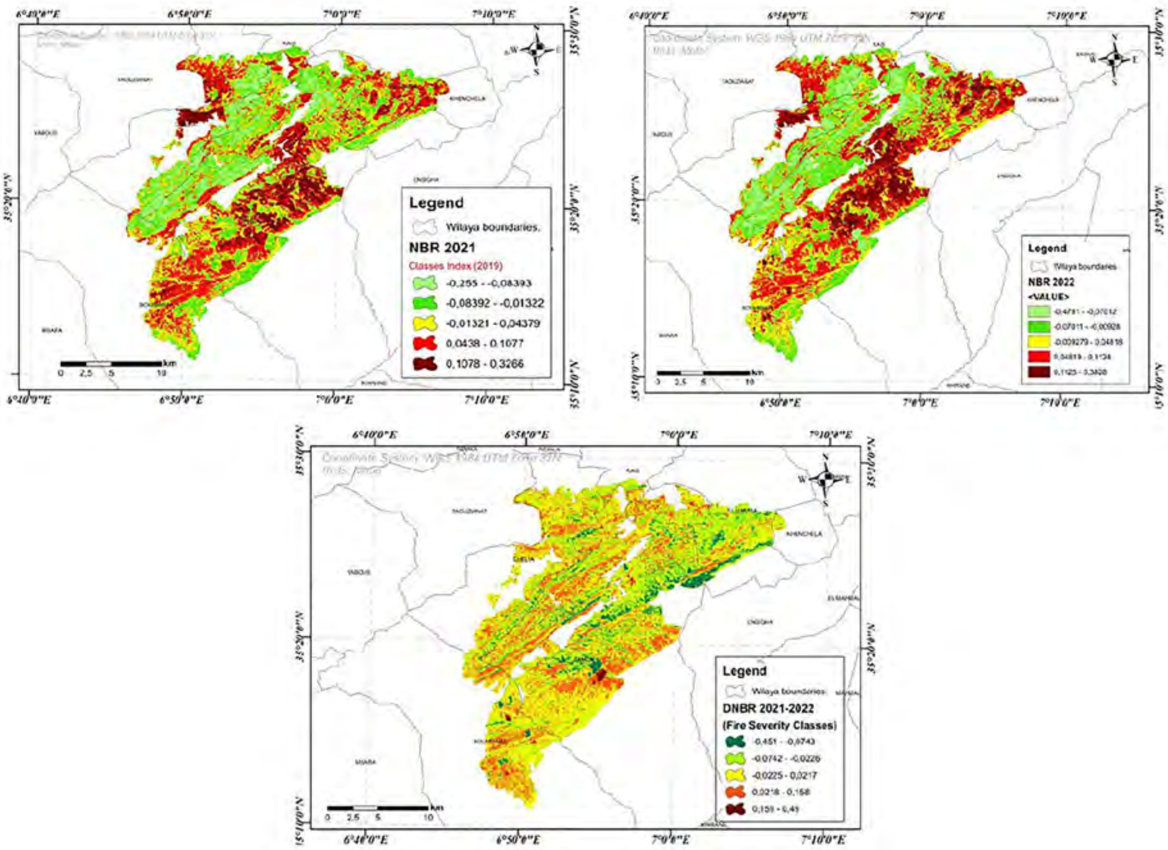

FIGURE 14
Spatial Distribution of Burn Severity in the Study Area Based on dNBR (2021–2022).

TABLE 12 DNBR classes (2020–2021) according to burn severity.

DNBR class (2020–2021)	Area (ha)	Area (%)	Burn severity
[-0.579, −0.171]	1357.06	4.97	Very low
[-0.170, −0.0354]	5818.70	21.31	Low
[-0.0353, 0.0292]	11670.16	42.74	Moderate
[0.0293, 0.113]	7066.53	25.88	High
[0.114, 1.07]	1392.56	5.10	Very high

TABLE 13 DNBR classes (2021–2022) according to burn severity.

DNBR class (2021–2022)	Area (ha)	Area (%)	Burn severity
[-0.4718, −0.1218]	232.09	0.85	Very low
[-0.1217, −0.05315]	589.79	2.16	Low
[-0.05316, 0.1274]	5938.84	21.75	Moderate
[0.1275, 0.2175]	13982.89	51.21	High
[0.2176, 0.8804]	6558.66	24.02	Very high

identified unburned or weakly affected vegetation. The dNBR results further refined this pattern by distinguishing a gradient from unburned or recovering areas to moderate, high, and very high burn-severity classes (Tables 12, 13). The predominance of moderate-to-high dNBR classes indicates that the 2021 event caused extensive canopy damage and major structural alteration across large parts of the massif. This interpretation is consistent with recent studies showing that dNBR-derived fire inventories and severity gradients remain highly effective for post-fire analysis and for subsequent wildfire zonation workflows (Guria et al., 2025).

Ecologically, these severity classes are not merely cartographic categories. Moderate and high burn severity generally reflect strong reductions in live biomass, litter consumption, understory destruction, and increased exposure of mineral soil, all of which can reduce short-term regeneration potential and increase the risk of post-fire erosion. In Mediterranean forests, severe fire effects are often associated with delayed vegetation recovery and stronger changes in ecosystem functioning, especially where drought stress and shallow soils constrain regeneration (Cristal et al., 2025).

The spatial distribution of high-severity classes in the central and southern sectors of the massif suggests that fire behavior was reinforced by fuel continuity, topographic confinement, and adverse meteorological conditions during the 2021 summer season. This interpretation is consistent with recent studies showing that wildfire severity is strongly influenced by the interaction between vegetation structure, pre-fire condition, and fire-weather conditions, including drought and strong winds (Turco et al., 2017; Zerouali et al., 2025).

3.4.2 Ecological significance of burn severity classes

The prevalence of moderate-to-high severity classes indicates that the 2021 wildfire was not a superficial surface disturbance, but a major ecological event with likely consequences for forest resilience. In semi-arid forests, severe burning can degrade organic matter, weaken soil structure, and reduce seed-bank viability, thereby limiting natural regeneration and favoring secondary degradation processes. These effects are especially critical in pine- and cedar-dominated systems where post-fire recovery depends not only on species traits, but also on moisture availability, fire recurrence, and site conditions (Cristal et al., 2025).

3.5 Vegetation dynamics before and after fire

3.5.1 NDVI, EVI, and NDWI patterns

The temporal analysis of NDVI showed a clear decline in vegetation greenness during the 2021 wildfire year, followed by only partial recovery in 2022 (Figure 15). Before the fire, NDVI values indicated relatively healthy and spatially continuous vegetation cover in large parts of the massif. In contrast, the 2021 NDVI map displayed widespread reductions in greenness, especially in the areas identified as moderate-to-high severity by dNBR. In 2022, the burned sectors remained spectrally distinct, indicating that vegetation recovery had begun but was still incomplete (Figure 16).

This pattern is ecologically meaningful. In Mediterranean ecosystems, post-fire vegetation recovery is rarely uniform during the first year after burning. Recovery trajectories depend on fire severity, topographic setting, fuel structure, and pre-fire vegetation vigor. Recent Sentinel-2 time-series work has

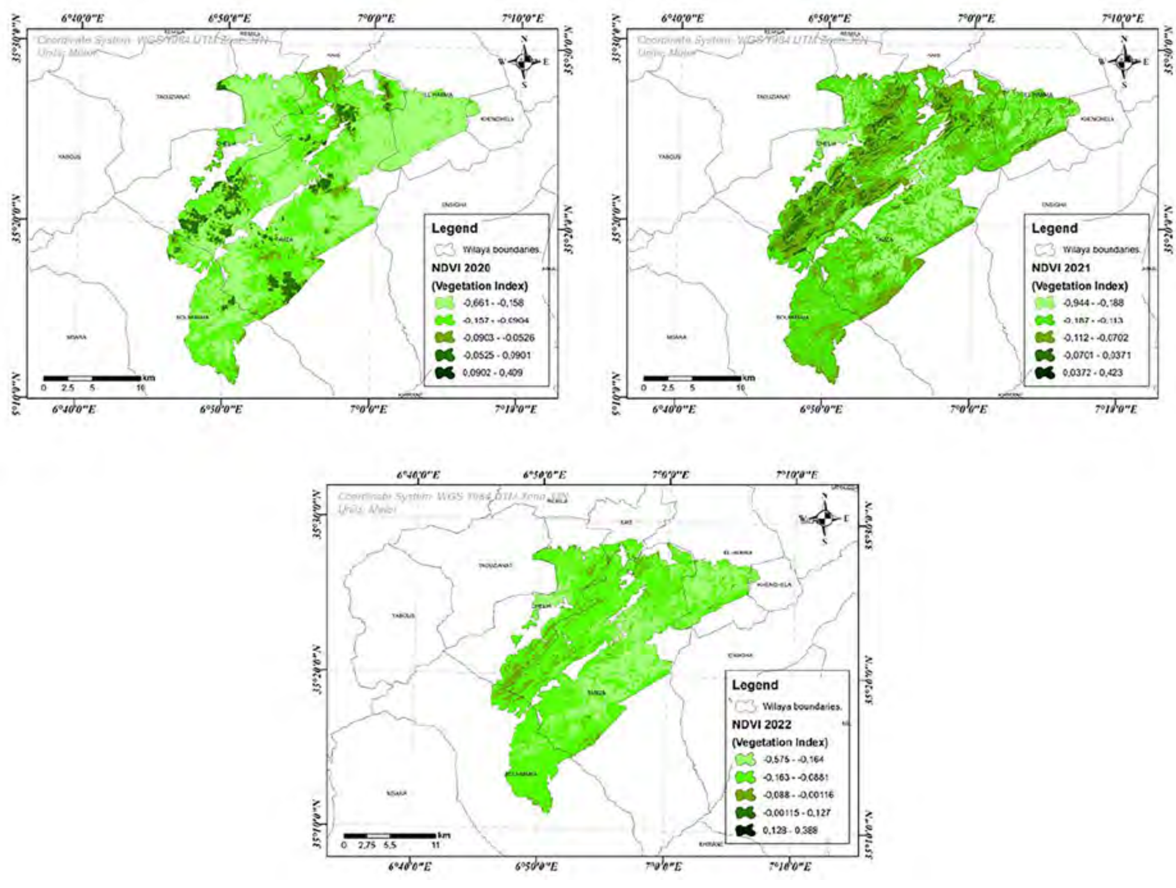

FIGURE 15
Spatio-temporal distribution of NDVI values in the study area for 2020, 2021, and 2022, illustrating vegetation dynamics before, during, and after the wildfire event.

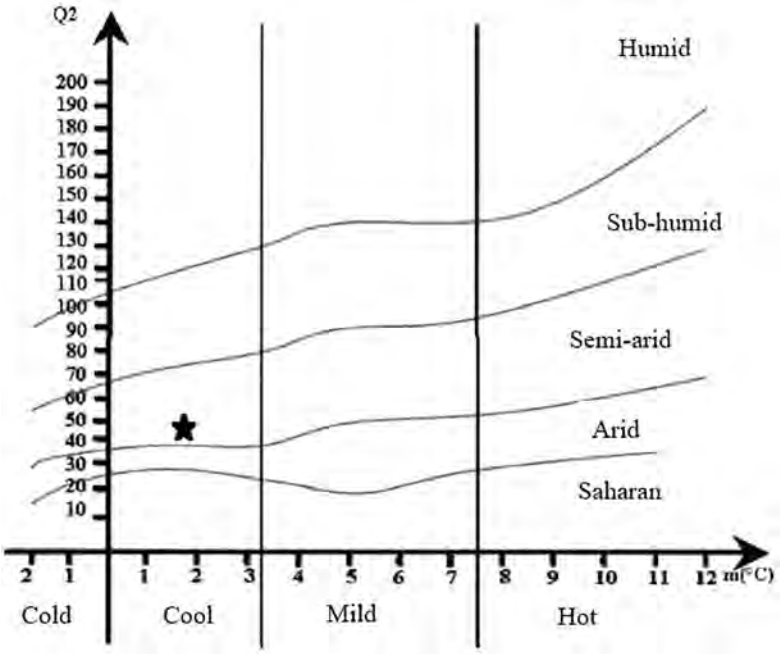

FIGURE 16
Emberger climagram of the Ouled Yagoub forest, showing the bioclimatic classification based on temperature and precipitation data for the period 2011–2020. The forest falls within the semi-arid stage with cool winters. * represents the climatic position of the Ouled Yagoub State.

shown that short-term post-fire recovery in Mediterranean vegetation is highly variable in space, and that recovery remains incomplete for a considerable period in severely burned stands (Zabeo et al., 2025).

EVI showed a similar temporal pattern but provided complementary information in denser vegetation, where NDVI may saturate. NDWI further indicated a decline in canopy moisture in burned and stressed sectors, confirming that post-fire vegetation condition was shaped not only by biomass loss, but also by reduced water content. In semi-arid environments, this combination of low greenness and low canopy moisture is particularly important because it signals both combustion impact and limited short-term recovery potential (Zabeo et al., 2025).

3.5.2 Recovery interpretation

The 2022 increase in NDVI and EVI values in some low- and moderate-severity sectors suggests an initial regeneration response, probably linked to herbaceous regrowth, resprouting, or partial canopy recovery (Figure 17). However, the persistence of low values in high-severity areas indicates that recovery remained spatially uneven. This agrees with Mediterranean post-fire studies showing that recovery may be relatively rapid in less severely affected areas but much slower where fire severity is high and site conditions are restrictive (Zabeo et al., 2025).

3.6 Climatic and environmental drivers of fire behavior

The exceptional severity of the 2021 wildfire can be largely attributed to unfavoable meteorological conditions, including high summer temperatures, prolonged drought, and strong winds reaching up to 60 km h1. These conditions significantly reduced fuel moisture content and enhanced fire spread and intensity. Wind-driven fire behavior played a critical role in determining the direction and speed of wildfire propagation, as evidenced by the alignment of severely burned areas with dominant wind directions. In Mediterranean and semi-arid regions, climate change is expected to exacerbate fire risk by increasing the frequency and intensity of heat waves and drought periods. The findings of this study are therefore consistent with broader regional trends indicating an escalation of wildfire hazards under changing climatic conditions.

4 Comparative performance of spectral indices

4.1 Indices detecting abrupt wildfire damage

Among the selected indices, NBR and dNBR were the most effective for detecting abrupt fire-induced vegetation loss and for mapping the severity gradient of the 2021 event. Their response is directly linked to the reduction of near-infrared reflectance and the increase of shortwave infrared reflectance after combustion. In the present study, these indices were more spatially coherent with the recorded burned area than the general greenness indices, confirming their central role in burn-severity assessment. This is fully consistent with recent remote-sensing studies, where dNBR remains a core indicator

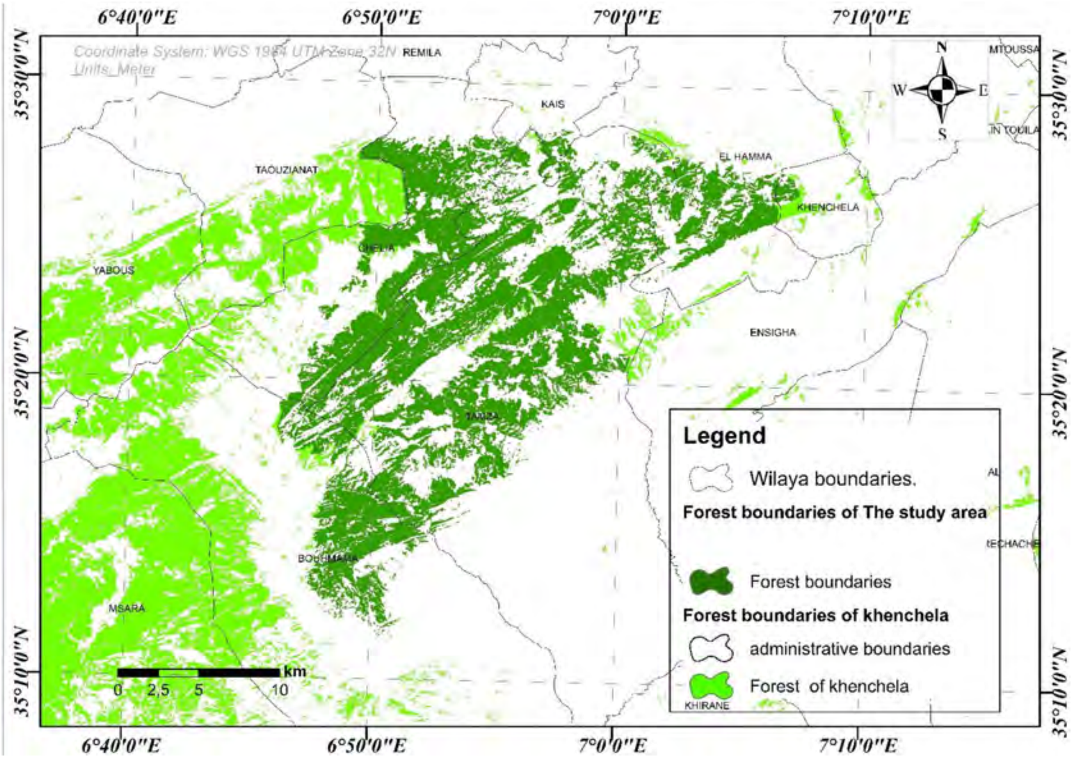

FIGURE 17
Forest boundaries of the study area.

for post-fire severity mapping and a useful basis for subsequent fire-probability and susceptibility analyses (Guria et al., 2025; Zerouali et al., 2025).

The soil-adjusted indices SAVI, MSAVI, and OSAVI complemented this interpretation by improving the detection of disturbed surfaces in areas where vegetation cover was sparse and soil exposure increased after fire. Their usefulness is particularly relevant in semi-arid landscapes, where bare soil can strongly influence spectral response and complicate the interpretation of post-fire vegetation condition (Zabeo et al., 2025).

4.2 Implications for forest management and DFCI strategies

The integration of spectral indices and spatial analysis provides valuable information for post f ire management and decision-making. The dNBR maps allow for the identification of priority areas for ecological restoration, soil stabilization, and long-term monitoring. Areas affected by high to very high burn severity should be targeted for immediate intervention to prevent soil erosion, biodiversity loss, and secondary disturbances. Furthermore, the results highlight the importance of strengthening Forest Fire Defense and Control (DFCI) strategies, including early warning systems, fuel management, and public aware ness programs. Remote sensing-based monitoring, particularly through platforms such as Google Earth Engine, offers a cost-effective and scalable approach for continuous fire risk assessment and rapid post-fire evaluation in forested landscapes.

4.3 Vegetation resilience and dieback processes

The combined analysis of vegetation indices provided valuable insights into post-disturbance vegetation resilience and the differentiation between wildfire-induced damage and dieback processes. Vegetation resilience was primarily assessed through the temporal evolution of NDVI and EVI values following the 2021 wildfire event. Areas affected by low to moderate burn severity exhibited a progressive increase in vegetation greenness during the post-fire period, indicating a relatively high capacity for natural regeneration. This response is characteristic of Mediterranean ecosystems, where many species possess adaptive strategies such as resprouting and seed bank activation. In contrast, areas classified as high to very high burn severity showed persistently low NDVI and EVI values, suggesting limited short-term recovery and reduced ecosystem resilience. These zones are likely to experience delayed regeneration, increased soil exposure, and heightened vulnerability to erosion and secondary disturbances. Dieback processes were identified based on a gradual decline in vegetation indices without corresponding high dNBR values. In these areas, NDVI and GCI exhibited progressive reductions over time, reflecting chronic vegetation stress rather than abrupt biomass loss. Additionally, NDWI values indicated a significant decrease in vegetation water content, highlighting the role of prolonged drought and water stress in driving dieback dynamics. The distinction between wildfire impacts and dieback processes is clearly captured by spectral behavior. Wildfire-affected areas are characterized by abrupt decreases in NDVI and strong positive dNBR values, reflecting combustion-driven biomass loss. Conversely, dieback areas show a smoother temporal decline in vegetation indices, near-zero dNBR values, and increasing pigment stress indicators such as SIPI. This spectral differentiation is essential for accurate in interpretation of vegetation degradation processes and for avoiding confusion between fire-induced damage and climate-driven vegetation decline.

Overall, the integration of multiple spectral indices enabled a robust discrimination between wildfire effects and dieback processes, providing critical information for post-fire management, restoration planning, and long-term monitoring of ecosystem resilience under changing climatic 17 conditions.

4.4 Limitations and future research

This study has several limitations that should be considered. First, no comprehensive ground-based validation dataset was available, which limited the direct verification of burn-severity classes and vegetation-condition patterns derived from spectral indices. Therefore, the present assessment should be interpreted as ecological and semi-quantitative rather than as a fully validated inferential analysis. Second, the relatively limited temporal window restricts the evaluation of longer-term post-fire vegetation dynamics and recovery processes. In addition, the framework relied mainly on spectral responses without explicitly integrating climatic, edaphic, or other environmental variables that may influence post-fire ecosystem condition.

Future research should incorporate representative field data from burned, unburned, and dieback-affected areas to improve validation and strengthen ecological interpretation. Longer temporal series should also be considered to better capture recovery trajectories and delayed vegetation responses after fire. In addition, the integration of climatic, topographic, and soil-related variables, together with advanced approaches such as machine-learning classification and multi-source data fusion, would improve the robustness and transferability of post-fire assessment in semi-arid forest ecosystems.

5 Conclusion

This study demonstrated that a multi-index remote-sensing framework can improve the distinction between abrupt wildfire-induced damage and progressive forest dieback in the Ouled Yagoub State Forest. In relation to the study objectives, the results showed that NBR and dNBR were the most effective indices for mapping burn severity, NDWI captured rapid post-fire canopy moisture loss, and SIPI, supported by GNDVI and GCI, was more informative for identifying progressive physiological stress associated with dieback. NDVI and EVI provided complementary information on overall vegetation decline and short-term post-fire recovery.

The findings also confirmed that the 2021 wildfire was an exceptional event, with 6,422.558 ha burned, and that moderate burn severity dominated large parts of the massif. The persistence of low vegetation and moisture values in several sectors one year after the fire indicated incomplete short-term recovery and highlighted differences in resilience across the study area.

Beyond its analytical value, the proposed multi-index framework provides a practical basis for operational wildfire monitoring, post-fire impact assessment, dieback detection, and restoration planning in semi-arid forest ecosystems. By combining fire-sensitive, moisture-related, and pigment-stress indices, the approach supports more targeted management decisions in environments where wildfire effects and chronic vegetation decline may occur simultaneously. Future work should strengthen this framework through field validation, longer time-series analysis, and the integration of climatic and topographic variables.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary material, further inquiries can be directed to the corresponding authors.

Author contributions

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/ffgc.2026.1809572/full#supplementary-material

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